Subject : Biçimsel Diller ve Otomata Teorisi

Date :...../...../.....

Otamat Teorisi (Automata Theory) Nedir

Otomata teorisi (özdevinim kuramı ya da otomata teorisi) teorik bilgisayar biliminde soyut makineleri (matematiksel makine ve sistemler) ve bu makineleri kullanarak hesaplama problemlerinin çözülebilmesini araştıran dalıdır. Bu soyut makinelere otomat denir.

Terminoloji

Alfabe (Alphabet) : Σ sonlu olan ve boş olmayan bir dizi semboldan oluşan bir kümedir

$\Sigma = \{0,1\}$ Binary (ikili) Alfabe

$\Sigma = \{a,b,c,\ldots z\}$ küçük harflerden oluşan alfabe

String : Alfabe'den seçilen harflerin birleşiminden oluşan yazılışlara string denir. Boş string ε (epsilon) sembolü ile gösterilir
Stringlerin uzunluğu yanlarındaki iki çizgi ile gösterilir

$|abzdet| = 5$ $|b| = 1$ $|\varepsilon| = 0$

Alfabenin kuvveti : Alfabeler sadece tek karakter veya semboller den oluşuyor. Alfabenin k'ıncı kuvvetini alarak k uzunluğunda alfabeler elde edebiliriz

$\Sigma = \{0,1\}$ $\Sigma^0 = \{\varepsilon\}$ $\Sigma^1 = \{0,1\}$ $\Sigma^2 = \{00,01,10,11\}$

$\Sigma^2 = (0,1) \times (0,1) = 00,01,10,11$

İki Stringi birleştirmek (concatenation)

↳ Elimizde a ve b olmak üzere iki tane string olsun

$a = xyz$ $b = wy$ $ab = xyzwy$

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Dil (Language) : Bir alfabenin tüm kuvvetlerinin oluşturduğu kümenin içinden seçilen belirli bir string setine dil denir

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$$

- Boş küme bir boş dildir. Pki: (Φ) sembolü ile gösterilir. Φ
- $\{\varepsilon\}$ bu bir dildir ve boş dil ile aynı şey değildir
- Türkçe ve İngilizce bir dildir

Set Former : Dilleri göstermek için yaygın bir yöntemdir

Set-former gösterim şekli: {a | a hakkında bir açıklama}

{a | a sadece 0 ve 1 lerden oluşur}

{a | a sadece çift sayılardan oluşur}

$\{a^n b^n \mid n > 0\}$

Karar Problemi (Decision Problem)

Bir stringin verilen bir dil içinde olup olmadığını anlamaya karar problemi denir. Girdi boyutuna göre çok büyük bilgisayar kaynakları gerektirebilir. Bu problem hiç çözülemeyen bir problem bile olabilir.

Bu problemi çözmek için otomata teorisi kullanılır.

Deterministik otomatlar : Bir stringi okurken gelen her karakter için gideceği tek bir yön veya hiç yön olmayan otomatlardır.

Deterministik Olmayan Otomatlar: Bir stringi okurken gelen her karakter için gideceği birden fazla yön olan otomatlardır.

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3

State (Q) : her yuvarlak bir state'dir

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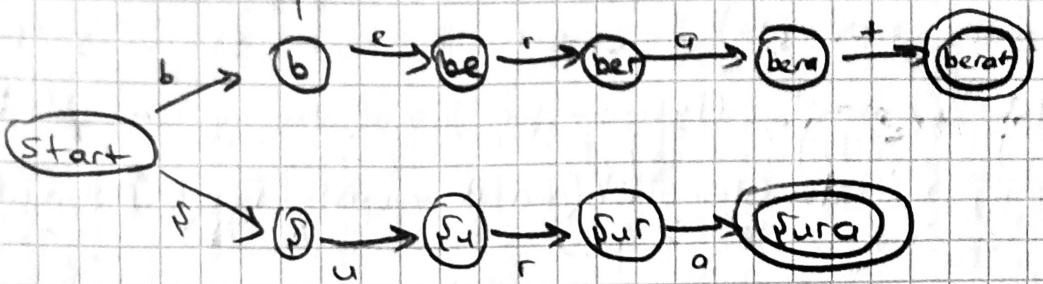
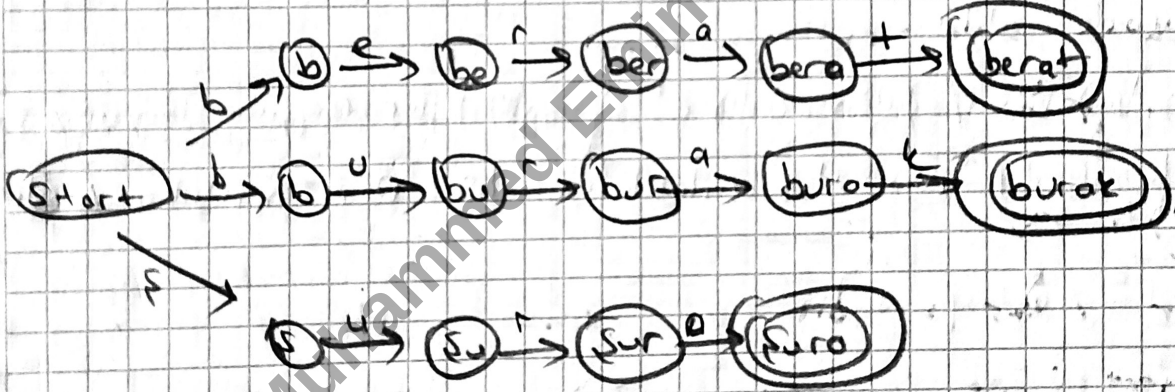
Örnek: Deterministik Otomat → ör : berat



Verilen string burak olsaydı b'den sonra tıkanıcağı için death state durumu duyar bu otomat bitmediği için kabul edilmiyor çünkü stringin sonuna kadar gelmesi lazım

Ör: Deterministik Olmayan Otomat :



burak stringi için b olan yolların ikisinede giriyor tıkanandan devam etmiyor diğer yolu seçip stringin sonuna ulaşıyor, Böylece bu örnek bu dilde kabul edilen bir string oluyor.

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DFA (Deterministik Finite Automata)

Sonlu otomat (Finite Automata) Çeşitleri

- Deterministik Sonlu Otomatlar (Deterministic Finite Automaton DFA)
- Deterministic Olmayan Sonlu Otomatlar (Non-Deterministic Finite Automata NFA)

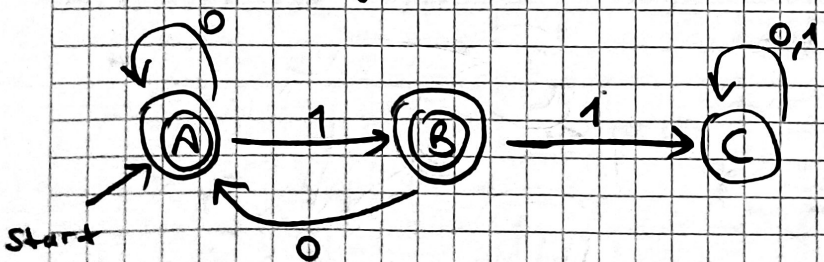
Deterministic Finite Automata (DFA)

- Q sonlu state'leri temsil eder. Genel olarak Q1, Q2, Q3 ---, QN
- Σ, alfabeyi temsil eder
- Delta bulunduğumuz state'de hangi yöne gideceğimizi belirler
- q0 genellikle başlangıç (start) state'i olur
- F, son state yani final state'i temsil eder. Final state iki tane çember ile ifade edilir.
- DFA tarafından kabul edilen dillere düzenli diller (regular languages) denir. Her düzenli dil için oluşturulabilecek bir DFA mutlaka vardır.

$A = (Q, \Sigma, \delta, q_0, F)$

Stateler $Q = (q_1, q_2, q_n)$ $\Sigma = \{0,1\}$ $\delta(q_1, 0) = q_2$

Ardışık 1 olmayan tüm stringleri kabul eden bir DFA



$Q = \{A, B, C\}$ $\Sigma = \{0,1\}$ $q0 = A$ $F = \{A, B\}$

ör: 000 1100
AAABC → burada C'ye gelince sonlandığı için bu dili kabul etmiyor

ör: 0 00 10010
AAA BAA BA
Burada final state {A, B} koşulunu sağladığı için bu dil kabul edilebilir.

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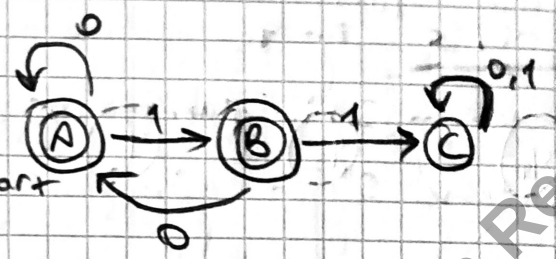
Örnek DFA, Delta Fonksiyon Gösterimi ile 0100 girdisini okuyalım



$\delta(A,0) = A$

$\delta(A,1) = B$

$\delta(B,0) = A$

$\delta(A,0) = A$

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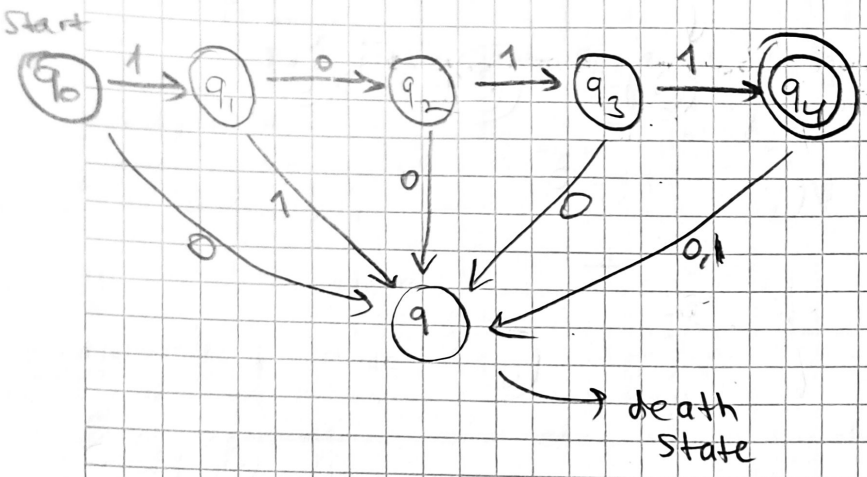
Teori-2

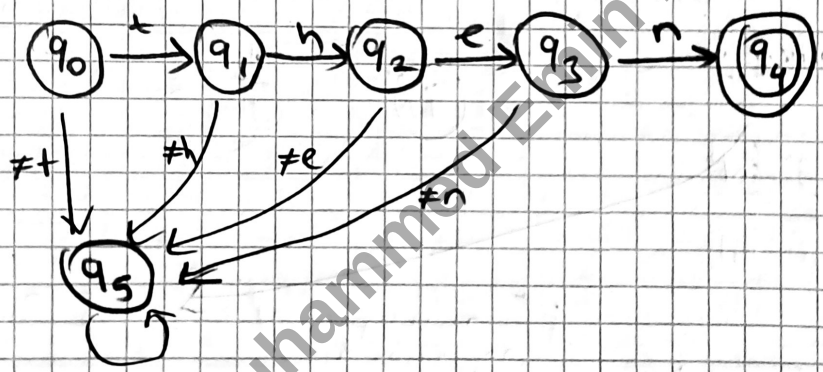
Subject :

Date :...../...../.....

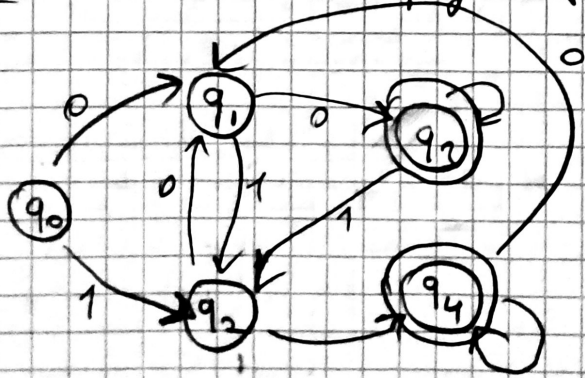
Ex: Build a deterministic DFA that accepts only the string 1011



Ex: Build a deterministic DFA that accepts only the word "then"



Ex: What is the language of this DFA



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Lab - 1

7

Subject :

Date :...../...../.....

- DFA, regular language' dir.

Ex:1 : Assume you have this alphabet $\Sigma = \{b, m, a\}$

$\Sigma^1 = \{b\}, \{m\}, \{a\}$

$\Sigma^2 = \{b,m\}, \{b,a\}, \{m,a\}$ ———

$\Sigma^3 = \{b,b,b\}, \{b,m,a\}, \{b,m,m\}$ ———

$\Sigma = (\Sigma^1 \cup \Sigma^2 \cup \Sigma^3$ ———

Ex:2 / Assume you have this alphabet $\Sigma = \{a,b\}$ give to examples of different languages that can be generated this alphabet

Ex:3 / The set of strings over $\Sigma = \{a,b,c\}$ having bc substrings

abc, bca, ~~cab~~

Ex4/ $\{xacy \mid x,y \in \{a,b,c\}^* \}$ →

→ abcaaa ac a

ac is substring

a b ac a

x y

→ y'nin uzunluğu 4 olmak zorunda

Ex5/ $\{xay \mid x,y \in \Sigma$ and $|y|=4$

Kural :

xxx a yyyy → sağdan 5. harf a'dır

x'in uzunluğu herhangi olabilir

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Ex:6 / Desing DFA

$\Sigma = \{0,1\}$

$L = \{w \mid w$ contains an odd number of 0's$\}$

NOT

state => q_0, q_1 ---

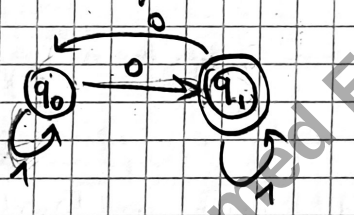
q_0 → initial state

q_f → final state (q_f)

$\delta(q_1, 0) = q_2$

Her adım belli (0,1)
1 yoldan sadece bir harf gelebilir

Çözüm → 0 sayısı tek sayı olmalıdır.



Transition Table

	0	1
q_0	q_1	q_0
q_1	q_0	q_1

100 → reject (en son q_0 00 bittiği için kabul edilmiyor)

0 → ✗
110 → ✗
101 → ✓
10001 → ✓

$\delta(q_1, 0) = q_0$

$\delta(\delta(\delta(q_1, 1), 1), 0) = q_0$

q_1

q_1

q_0

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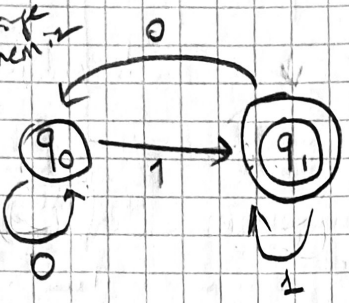
Ex: 7 / Design DFA

$\Sigma = \{0, 1\}$

$L = \{ w \mid w \text{ contains } 011 \}$

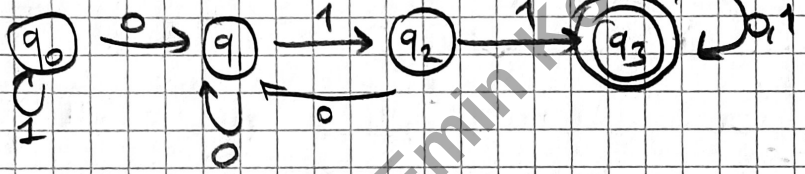
bu doğru garantiye çözmem ama alarak kazanım



	0	1
q_0	q_0	q_1
q_1	q_0	q_1

001
1001011 → ✓
→ ✓

Burda garantiye alıyor

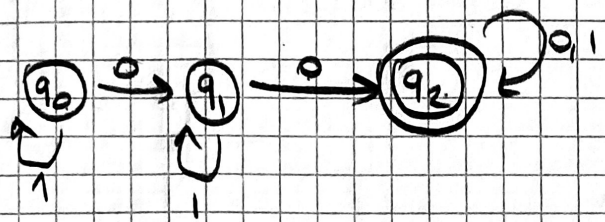


Ex 8 / Design DFA

$\Sigma = \{0, 1\}$

$L = \{ w \mid w \text{ ends with } 00 \}$

Node sayısına karar verirken iki garantiye almamız lazım



	0	1
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_2	q_2

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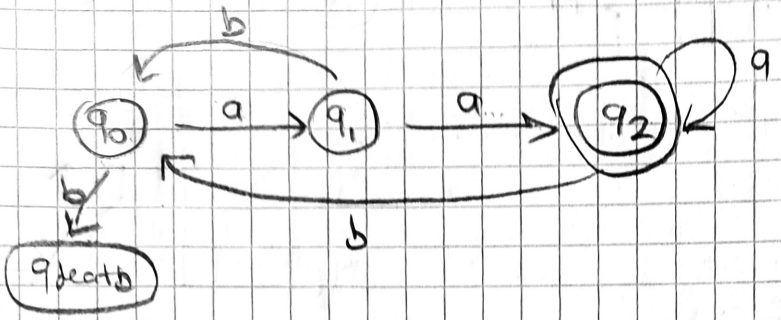
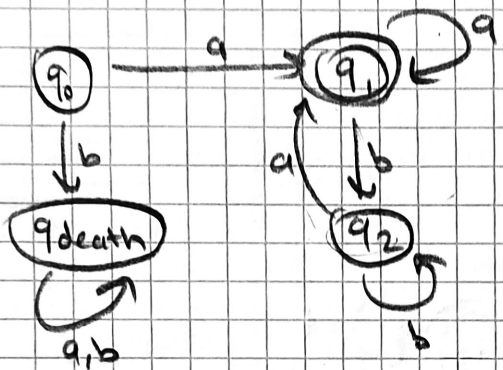
Ex/9 $\Sigma = \{a, b\}$

$L = \{w \mid w$ start with a and w end with a$\}$

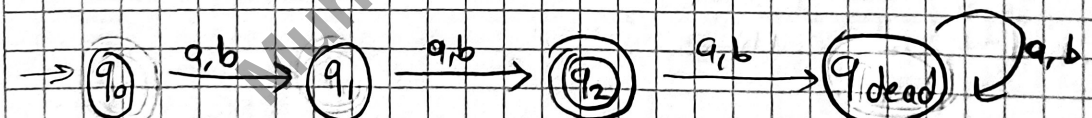


	a	b
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_0



Teori-3

Ex/1 : Construct a DFA that accepts set of all strings over $\Sigma = \{a, b\}$ of length at most 2 i.e. $|w| <= 2$



State	a	b
→ q_0	q_1	q_1
q_1	q_2	q_2
q_2	D	D
D	D	D

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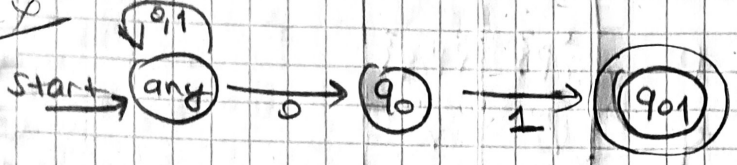
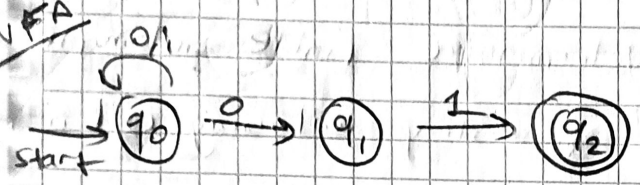
NFA (Non Deterministic Finite Automata)

Ex

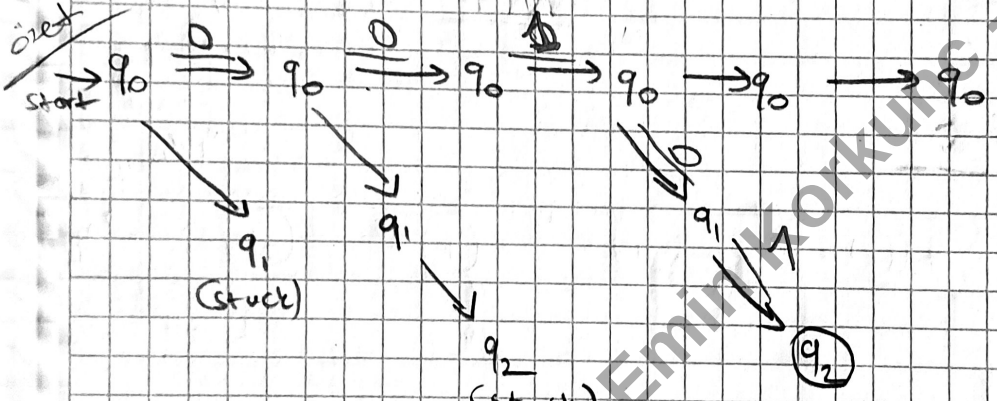


Accept all strings that end 01

NFA



özet



0 0 1 0 1

state	0	1
→ q_0	$\{q_0, q_1\}$	q_0
q_1	$\emptyset$	q_2
* q_2	$\emptyset$	$\emptyset$

* Final state

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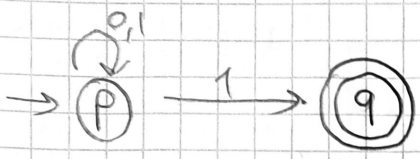
Ex 2 / NFA.



state	0	1
p	{p}	{p,q}
q	∅	∅

Ex 3 / Draw a deterministic and non-deterministic finite automate which accept 00 and 11 at the end of a string containing 0,1 in it.

DFA

NFA

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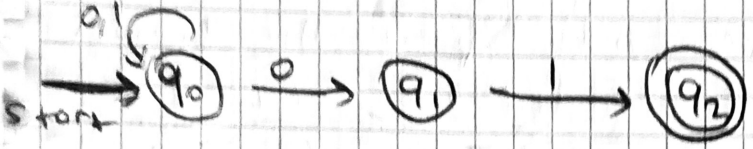
Subset Construction

2Q

Ex: Example for string end with 01



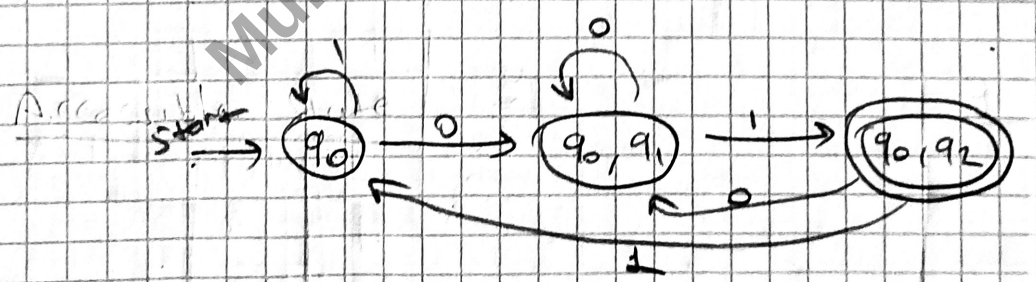
$2^3 = 8$

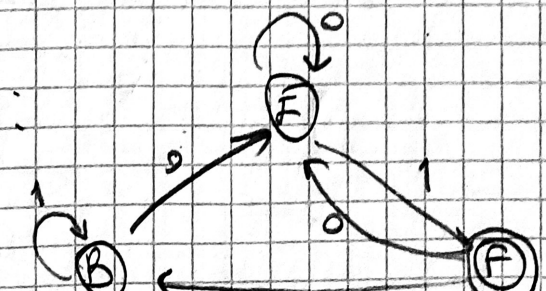
State	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$
$\{q_1\}$	$\{\emptyset\}$	$\{q_2\}$
$*\{q_2\}$	$\emptyset$	$\emptyset$
$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$*\{q_0, q_2\}$	$\{q_0, q_1\}$	$\{q_0\}$
$*\{q_1, q_2\}$	$\emptyset$	$\{q_2\}$
$*\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$

Rename →

	0	1
A	A	A
→ B	E	B
C	A	D
* D	A	A
E	E	F
* F	E	B
* G	A	D
* H	E	F

final state getince durabilir





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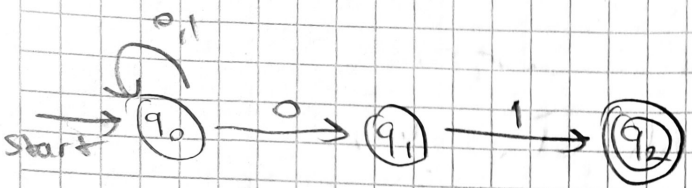
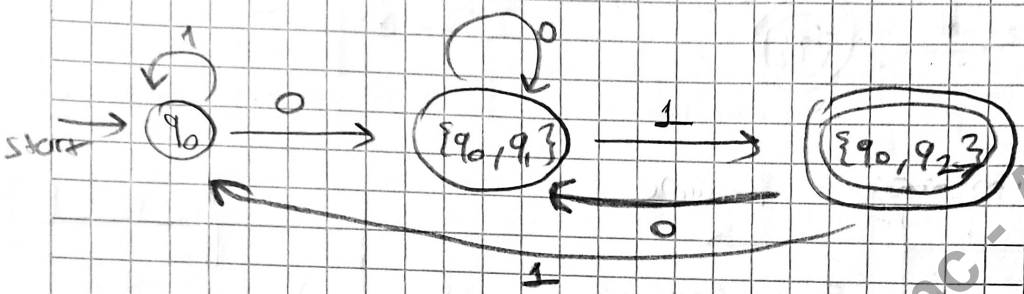
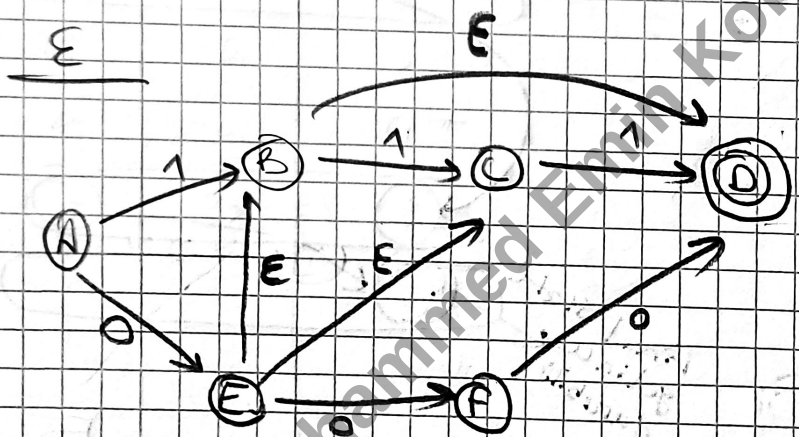
Accessible state

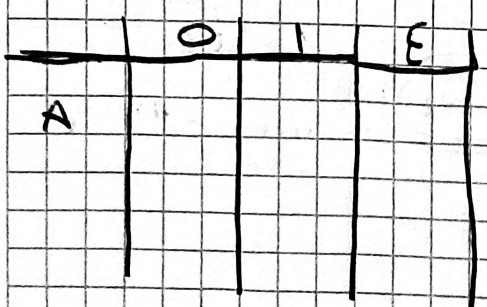
Lazy State





ε





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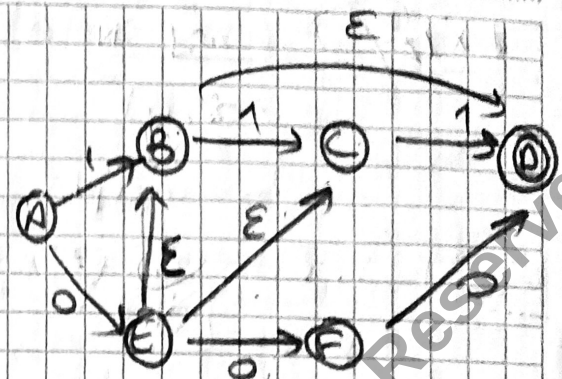
Closure of States

Ex/

$CL(A) = \{A\}$

$CL(E) = \{B, C, D, E\}$



↳ ε ile gidebilecegi yerler

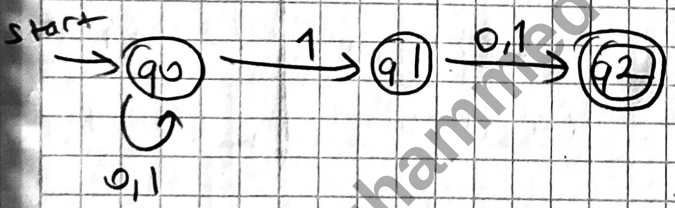
$\hat{\delta}(A, \varepsilon) = CL(A) = \{A\}$

$\hat{\delta}(A, 0) = CL(\{E\}) = \{B, C, D, E\}$

$\hat{\delta}(A, 01) = CL(\{C, D\}) = \{C, D\}$

LAB

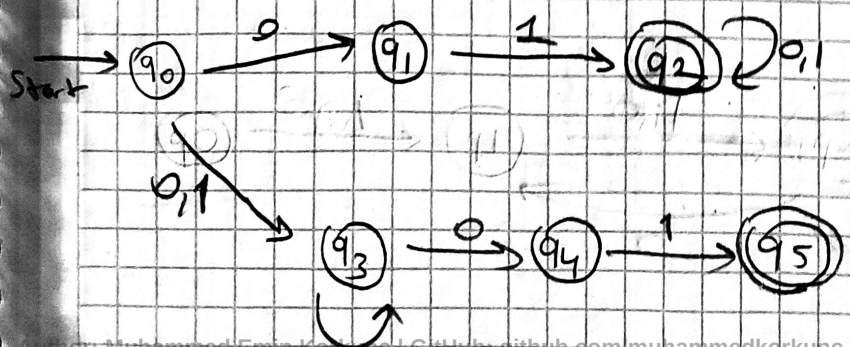
$\Sigma = \{0, 1\}$ NFA accepts — second character from right 0 1



Ex3 Design NFA $\Sigma = \{0, 1\}$

NFA accepts the string starts with 01 or end with 01 or a string containing 0,1

- 01 0101 00 but not 000111010



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Subject :

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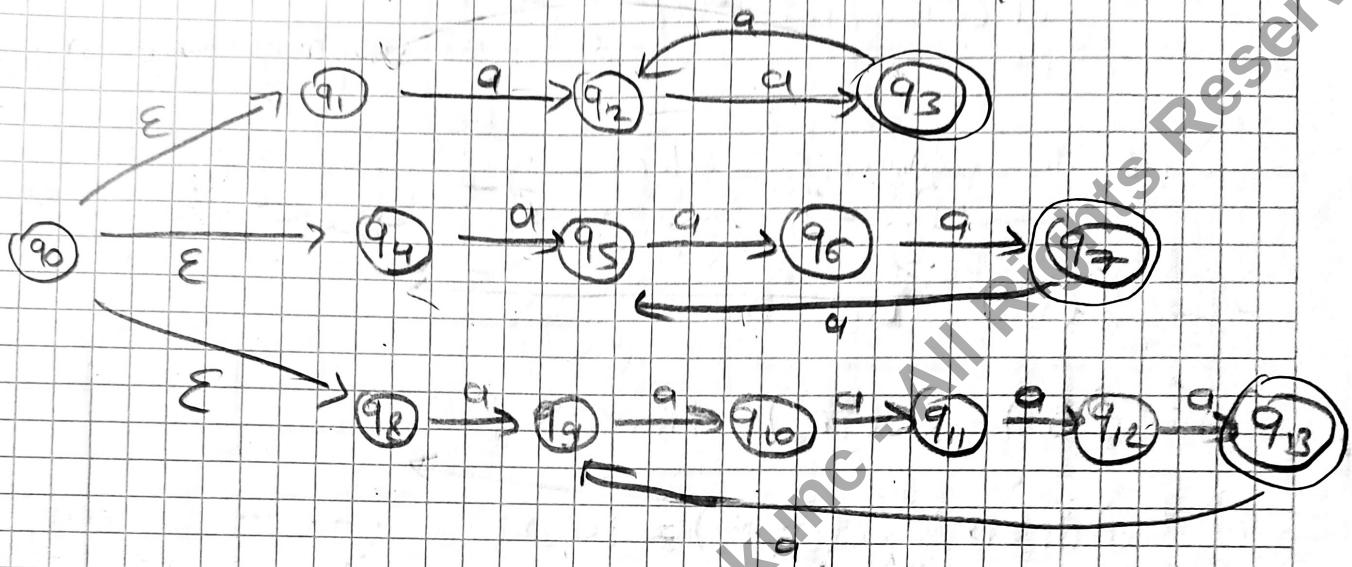
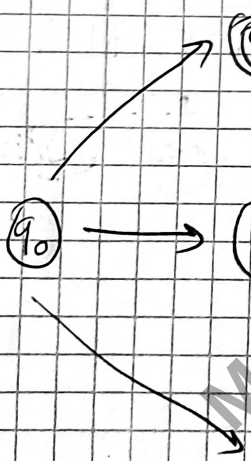
Ex 4/ Desing NFA

$\Sigma = \{a\}$

└ accepts strings of size multiples 2, 3 or 5





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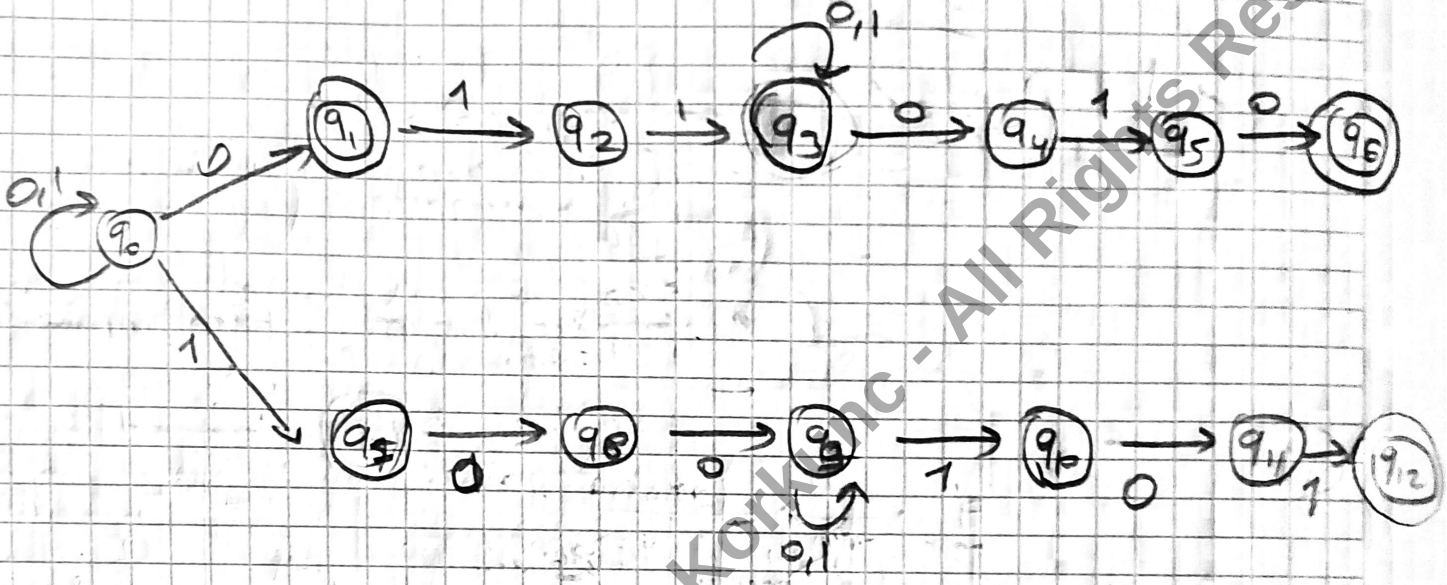
Ex 5/ Desing NFA

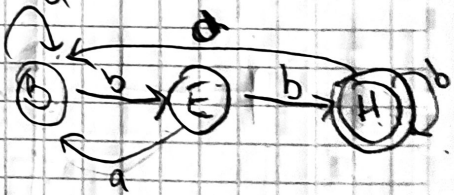
$\Sigma = \{0, 1\}$

NFA accepts end in 010 and have 011 somewhere preceding or end in 101 and have 100 somewhere preceding

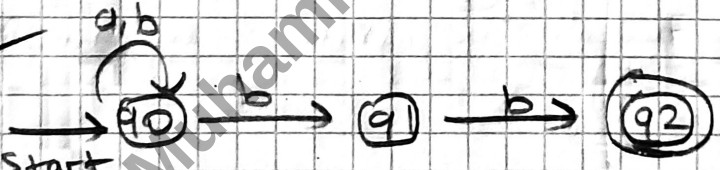


Ex 5

Convert NFA to DFA



NFA



Step 1: Subset construction $2^q = 2^3 = 8$

	a	b
$\{\emptyset\}$	$\emptyset$	$\emptyset$
→ q_0	q_0	$\{q_0, q_1\}$
q_1	$\emptyset$	q_2
★ q_2	$\emptyset$	$\emptyset$
q_0, q_1	q_0	q_0, q_1, q_2
★ q_0, q_2	q_0	q_0, q_1
★ q_1, q_2	$\emptyset$	q_2
★ q_0, q_1, q_2	q_0	q_0, q_1, q_2

Step 2 : rename

	a	b
→ A	A	A
B	B	E
C	A	D
★ D	A	A
★ E	B	H
★ F	B	E
★ G	A	D
★ H	B	H

A → $\emptyset$

B → q_0

E → q_0, q_1

H → q_0, q_1, q_2

q_1 → C

q_2 → D

F → q_0, q_2

G → q_1, q_2

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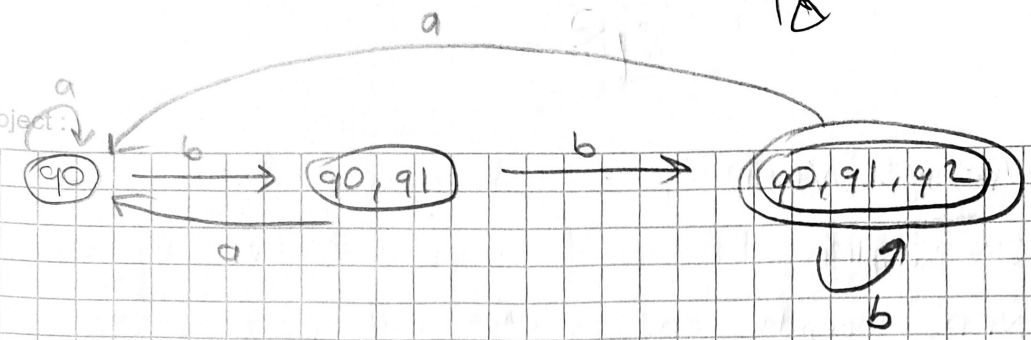

Ex : Convert following

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Subject :

Date : ..../..../.....

Sembol → a, b, 1, 2

Alfabe → {a, b}

String → aa, ab, c, ϵ

Language → L = {uzunluğu 2 olan stringler}
{aa, ab, ba, bb}

L = {Sonsuz sayıda string içeren bir language örneği}
{b ile başlayan stringler}

$\Sigma = \{a, b\}$

$\Sigma^0 = \epsilon$

$\Sigma^2 = \{a,b\}, \{a,b\} = \{aa, ab, ba, bb\}$ (concatenation = birbirine bağlama)

$\Sigma^3 = \{aaa, aab, aac \dots\}$

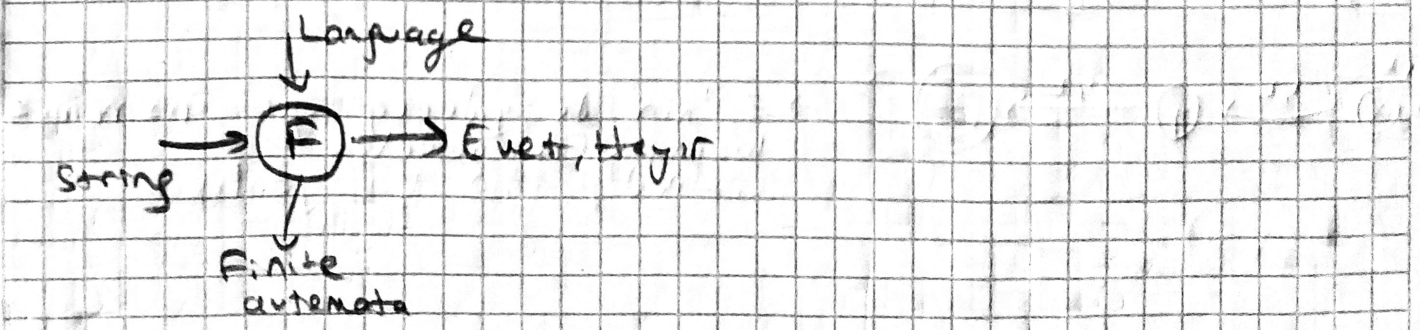
Σ^n = (n uzunluğunda bu alfabeyi kullanarak yapılabilecek stringler)
= aaa...a (n)

Ör: $\Sigma = \{a, b\}$

L = uzunluğu 2 olan stringler

L = {aa, ab, ba, bb}

String = {a, c}



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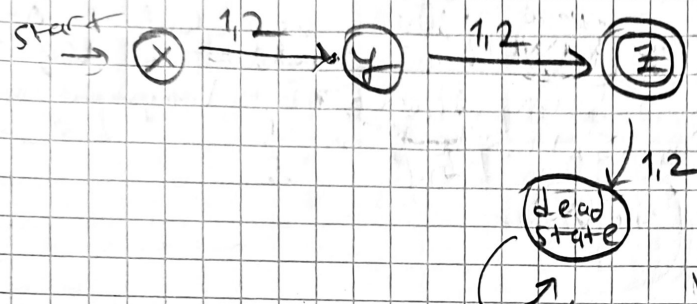
Subject :

Date :...../...../.....

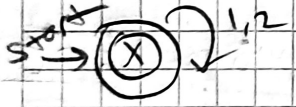
Ör: L uzunluğu 2 den stringler.
L kur. $\Sigma = \{1,2\}$ alfabesinde tanımlı. L 'i tanıyan DFA

Cevap → $L = \{11, 12, 21, 22\}$



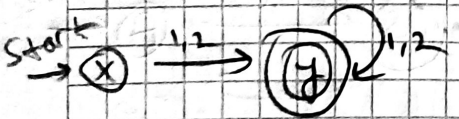
DFA 'da çalıştığımız için her durum verilen alfabeyi sağlamalıdır. yani Z 'de 1 ve 2 nin nereye gittiği gösterilmeli. Farklı bir state açsak string uzunluğu 3 durur bu nedenle dead state açtık

Hatalı durumlar

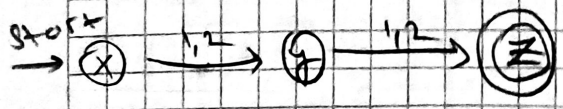


ϵ: $|\epsilon| \neq 2$
$|\epsilon| \neq 1$
$|\epsilon| = 0$ → epsilon uzunluğu sıfır

→ Bu nedenle bu şekilde çizince epsilon ϵ kabul edilmiyor. O nedenle şartları ihlal ediyor.



→ Burda gelen string 1 veya 2 olursa da cevap kabul ediliyor bu nedenle şartı ihlal ediyor



→ Z 'nin de gideceği yeri çizmemiz lazım DFA modelde çizmediği için bu yanlış

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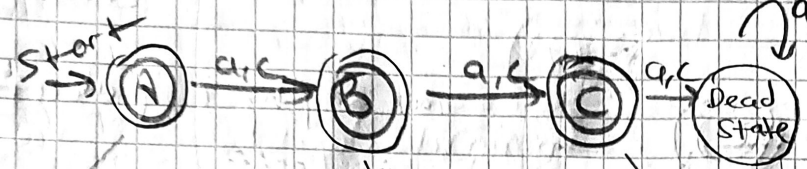
Subject :

Date :...../...../.....

Ör: Word $\in \{a,c\}$, $|word| \leq 2$, Bunu tanımlayan sonlu makine çiz.

Çözüm → $L = \{\epsilon, a, c, aa, ac, ca, cc\}$ (verilen alfabeden oluşabilecek dil)



Bu DFA yani sonlu makina olduğu için C'de nereye gittiğinde bilmeliyiz yani her state için alfabedeki her durumu kullanmalı

0 uzunluğundaki stringler

1 uzunluğundaki stringler

2 uzunluğundaki stringler

Uzunluk olarak 2'den küçükleri kabul ettiği için A final state olabilir

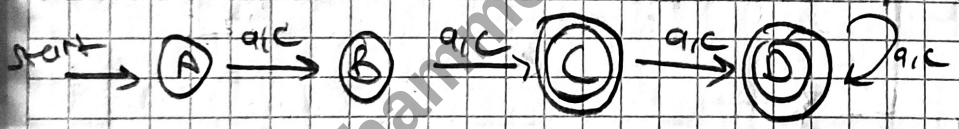
2'den küçük ve 2'ye eşit olan stringlerde final state olabilir

Dead state'in de alfabedeki durumlarda ne olduğunu bilmeliyiz Dead state sonsuz string alabilir fakat şart 2'den küçük veya eşit uzunlukta olduğu için final state olamaz

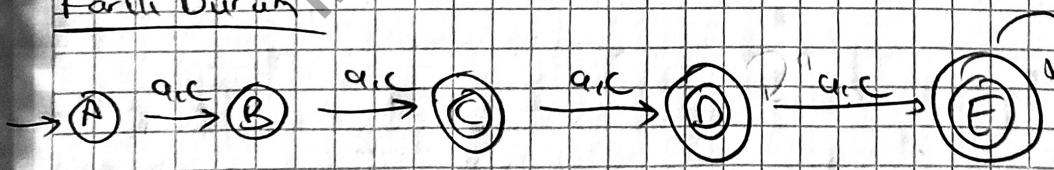
Ör:
Word $\in \{a,c\}$ $|word| \geq 2$ Bunu tanımlayan Sonlu makine çiz

Çözüm

$L = \{aa, ac, ca, cc, aaa, _____\}$



Farklı Durum



Bu çözümde doğru fakat buna minimal DFA yababiliriz üstteki çözümde olduğu gibi

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22 → Uzunluğunun %2 == 0 olan stringleri kabul eden DFA makinesi çizelim

DFA

Subject :

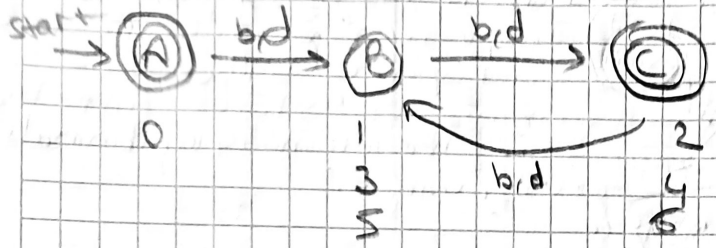
Date :...../...../.....

Ör: word ∈ {b,d}, |word| mod 2 = 0 , bu language için sonlu makine kur.

Çözüm

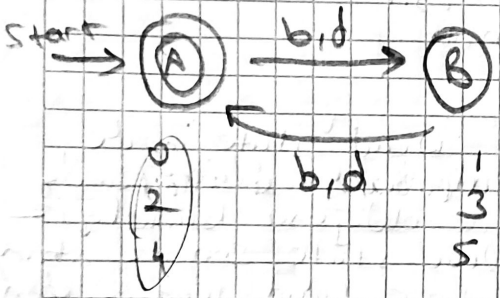
L = { ∈, bb, bd, bdbd ---- }

alfabe



Çözüm sağlıyor fakat minimal değil.

Makinede makinenin bazına kullanmak yanlış olduğu için cevabı minimal hale getirmeniz lazım

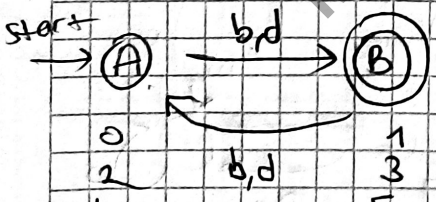


Cevap bu şekilde doğrudur minimal hale geldi.

Ör: word ∈ {b,d}, |word| mod 2 = 1 için DFA makinesi çiz

Çözüm

L = { b, d, bbd, dbb, ---- }



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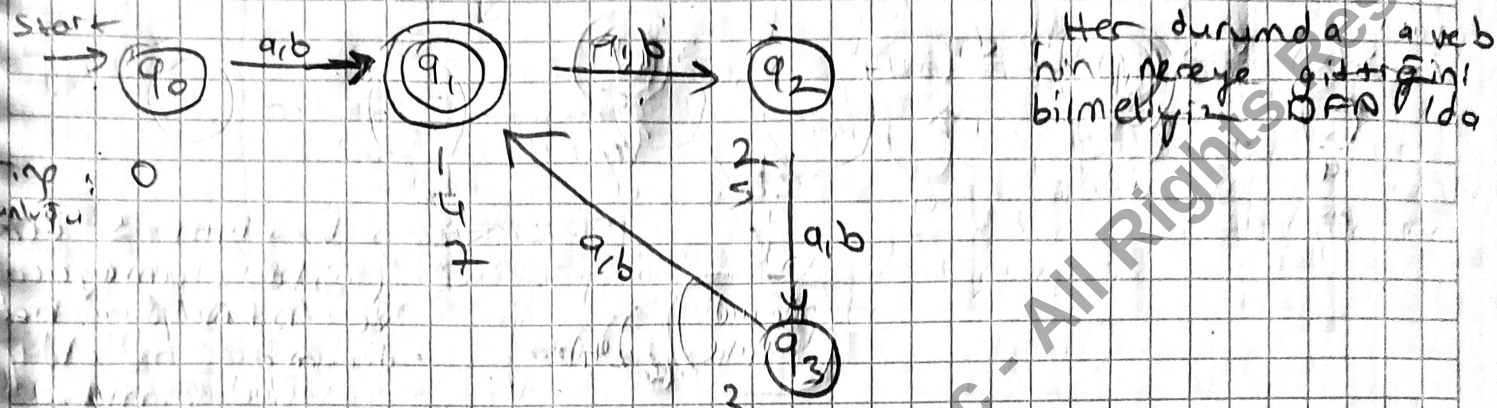
Subject :

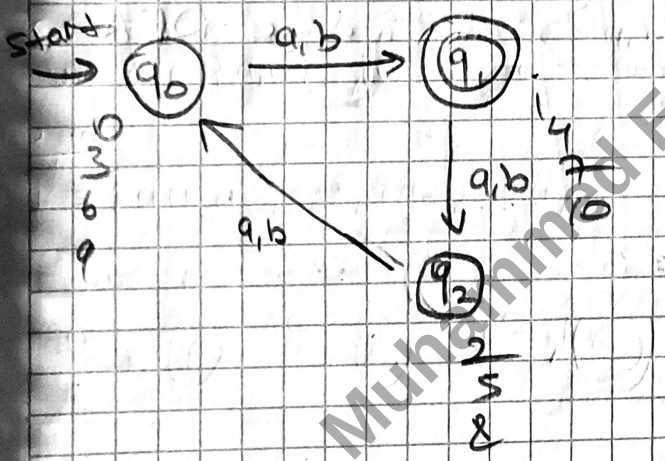
Date :...../...../.....

Ör: word ∈ {a,b} , |word| mod 3 = 1 , sonlu makine DFA çiz.

Çözüm: L = {a, b, aabb, aaab, ---



Her durumda a ve b'nin nereye gittiğini bilmeliyiz → DFA'da



Doğru cevap bu, çünkü daha minimal'dir.

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number b in string = 3

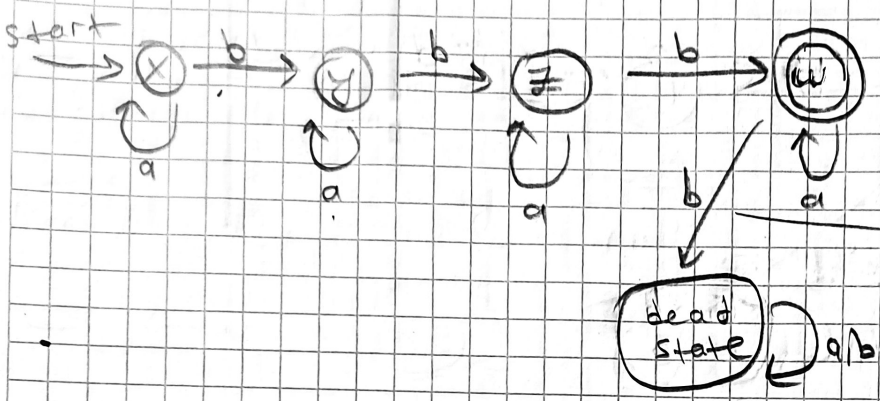
24

Subject :

Date :...../...../.....

Ör: word ∈ {a,b} , $n_b(word) = 3$ sonlu makine DFA çiz.

Çözüm → L = {bbb, bbba, babab, aaaaaaa bbb ...}

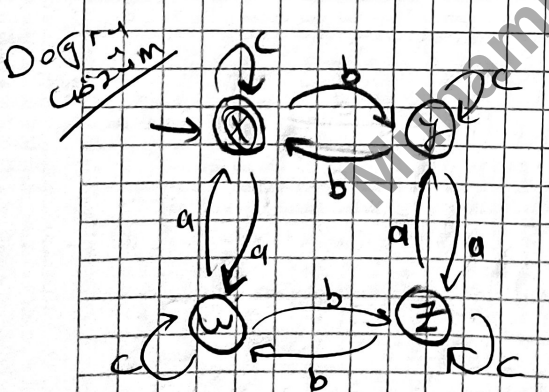


b sayısı 3'den fazla olamayacağı, ve DFA'da her durumda ne olduğunu göstermemiz için ölü durumu gösterdik.

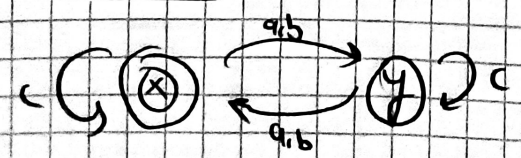
Ör: word ∈ {a,b,c} $n_b(word) \bmod 2 = 0$ $n_a(word) \bmod 2 = 0$ } sonlu makine DFA çiz.

Çözüm → L = {ϵ, bb, aa, aabb, bbbaa

Doğru Çözüm



Yanlış Çözüm



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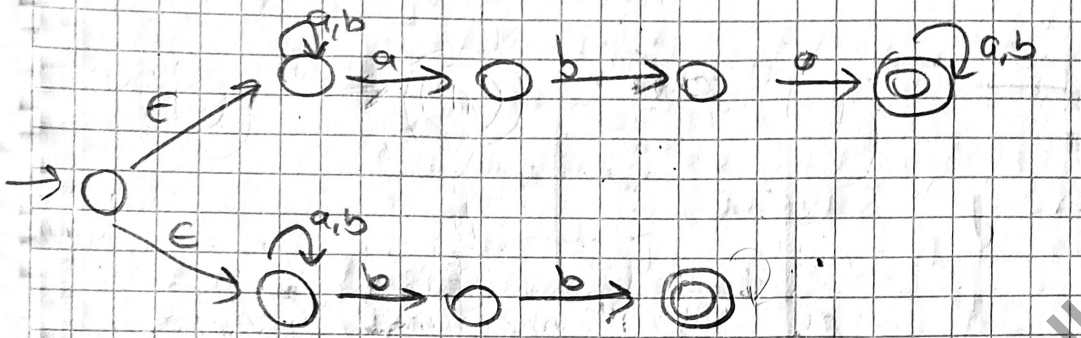
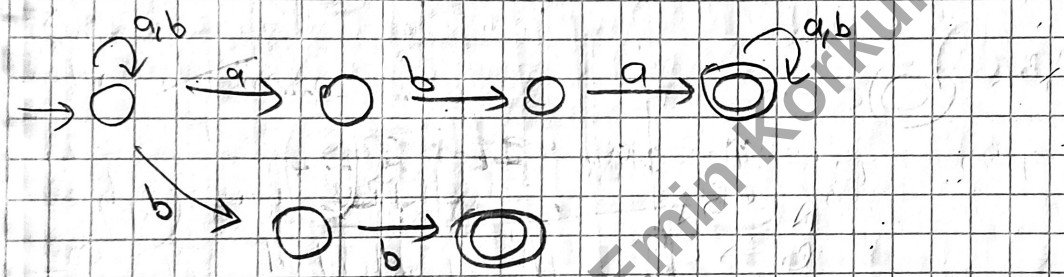
Subject :

Date :...../...../.....

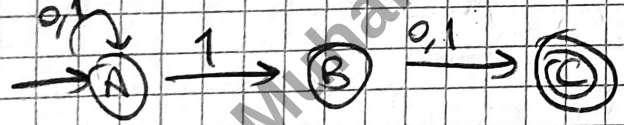
NFA Ex

Design NFA recognizes over the alphabet {a,b} that contain either the substrings "abb" or end with "bb".





Ex-2 : Convert the given NFA to DFA



Çözüm = (Subset Method) → $2^3 = 8$

	0	1
∅	∅	∅
→ A	A	{A,B}
B	C	C
* C	∅	∅
AB	{A,C}	{A,B,C}
* AC	A	{A,B}
* BC	C	C
* ABC	{A,C}	{A,B,C}

	0	1
$q_{\emptyset}$	$q_{\emptyset}$	$q_{\emptyset}$
→ q_1	q_1	q_4
q_2	q_3	q_3
* q_3	q_0	q_0
q_4	q_5	q_2
* q_5	q_1	q_4
* q_6	q_3	q_3
* q_7	q_5	q_7

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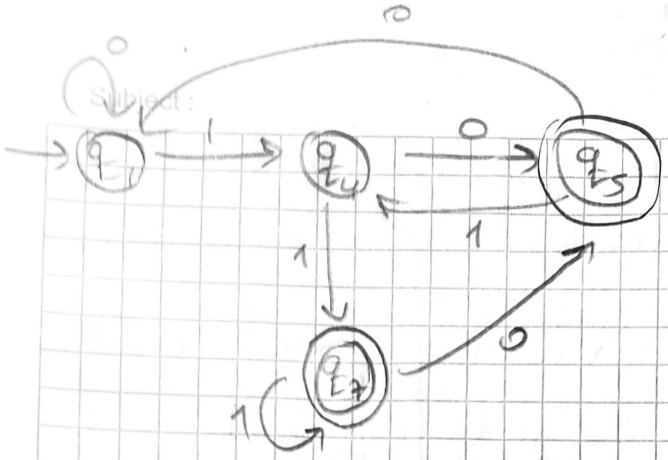
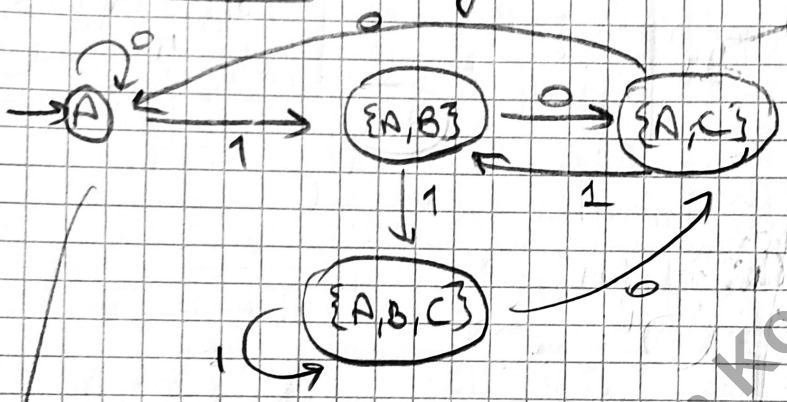
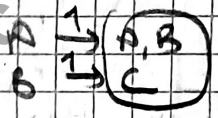
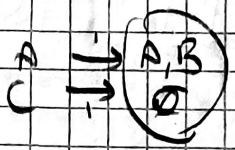
Date :...../...../.....

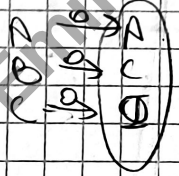


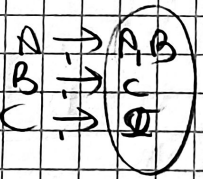
2. Çözüm : LAZY STATE











→ Başlangıçta NFA'de 1 iki konuma gidiyor
Fakat DFA'de her alfabe için bir duruma gidebilir
bunu düzenleyerek başlıyoruz

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Regular Expression

$L(E_1 + E_2) = L(E_1) \cup L(E_2)$

$L(E_1 E_2) = L(E_1) \cdot L(E_2)$ Concatenation

$L(E^*) = (L(E))^*$ Repeating

Or: RE's

$L(01) = \{0, 1\}$

$L(01+0) = \{01, 0\}$

$L(0(1+0)) = \{01, 00\}$

$L(0^*) = \{\epsilon, 0, 00, 000, \dots\}$

$L((0+10)^* (\epsilon+1))$ = all strings of 0's and 1's without two consecutive 1's

Ex

Write a regular expression for the set of string that consist of alternating 0's and 1's

$(01)^* + (10)^* + 0(10)^* + 1(01)^*$

Solution → $0(10)^*$ or $(01)^*0$ → 01010

$(01)^*$ → 0101010101

$(10)^*1$ or $1(01)^*$ → 1010101

$(10)^*$ → 10101

$\boxed{(01)^*} + (10)^* + \boxed{(01)^*0} + (10)^*1$

$(01)^*(\epsilon+0) + (10)^*(\epsilon+1)$

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$(\epsilon+1)(01)^*(\epsilon+0)$

Ör: Regular Expression

(a) The set of strings over alphabet $\{a,b,c\}$ containing at least one a and at least one b.

$(A+B+C)^*(A(A+B+C)^*B + B(A+B+C)^*A)(A+B+C)^*$

(b) The set of strings of 0's and 1's whose tenth symbol from the right end is 1.

$(0+1)^*1(0+1)^9$

(c) The set of strings of 0's and 1's with at most one pair of consecutive 1's

↳ En fazla yan yana 1 bulunma sayısı 1 'dir.

$(0+10)^*(11+\epsilon)(0+\epsilon)^*$

Ex

alphabet - $\Sigma=\{a,b\}$ → Başlangıç ne olursa olsun önemli değil

(a) All strings that do not end with aa

$\epsilon+a+b+(a+b)^*(ab+ba+bb)$

(b) All strings that contain an (even) number of b's → sıfır

$a^*(ba^*ba^*)^*$ ⇒ b'nin üstündeki repeat sembolü
$\epsilon, bb, bbbb, \dots$ → b'nin hep çift tekrar etmesini sağlar

(c) All strings which do not contain the substring ba

a^*b^*

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Ör: $\Sigma = \{0,1\}$ alphabet

Bu iki'si 11 gibi duyor

- Language of only two strings

Finite → (0+1)(0+1) or 00+01+10+11

→ (0+1)(0+1)(0+1) ? yanlış üç string var → hem ϵ hem boş eleman hem sonsuz tekrarı singeler

- Language of at least two strings, {00, 01, 10, 00, 000, ——

Infinite → (0+1)(0+1)(0+1)* → en az iki string en fazla sınırsız

→ (0+1)(0+1)(0+1)(0+1)* ? en az üç string

- Language of at most two strings

Finite → (0+1+ϵ)(0+1+ϵ) → bu değil

→ (0+1+ϵ)(0+1) ? → at most two without ϵ
or
at most two and at least one

- Language of even strings {00, 01, 10, ——, 0000, 0110}

Infinite → ((0+1)(0+1))*

(0+1)* ? → herhangi birşey olabilir
ϵ, 0, 11

- Language of odd string

Infinite → ((0+1)(0+1))* (0+1)

- what is the langugage of multiplies of 3 strings.

((0+1)(0+1)(0+1))*

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Subject : RE

Date : ...../...../.....

Ör : In $\Sigma = \{a, b\}$

- Strings that starts by a ?

  $a(a+b)^*$ → a ile başlar gerisi fark etmez

- String that end with b ?

  $(a+b)^*b$

- String containing a ?

  $(a+b)^* a (a+b)^*$

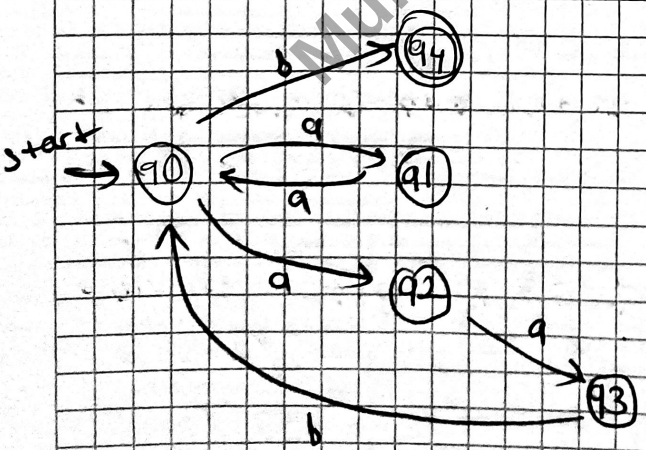
- Start and end with different symbols

  $a(a+b)^*b$ or $b(a+b)^*a$

- Start and end with same symbol

  $a(a+b)^*a$ or $b(a+b)^*b$ or a or b or ϵ

NFA and RE



$(aa)^*b + (aaab)^*b$

$= ((aa)^* + (aaab)^*)^*b$

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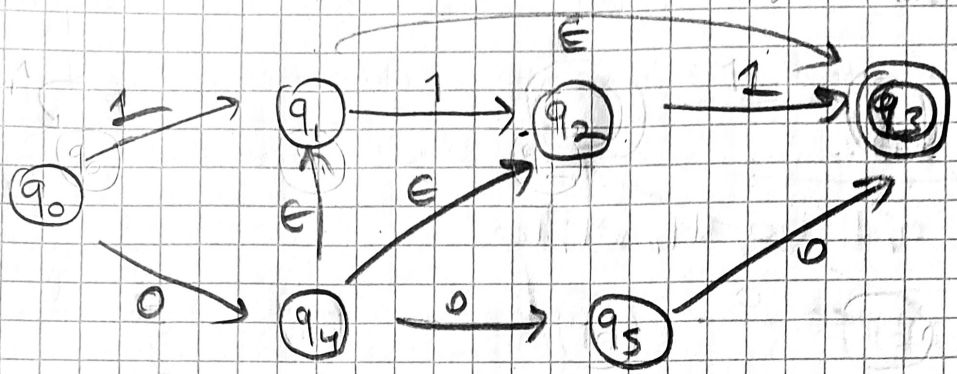
Subject : . Date :...../...../.....

Ex : NFA

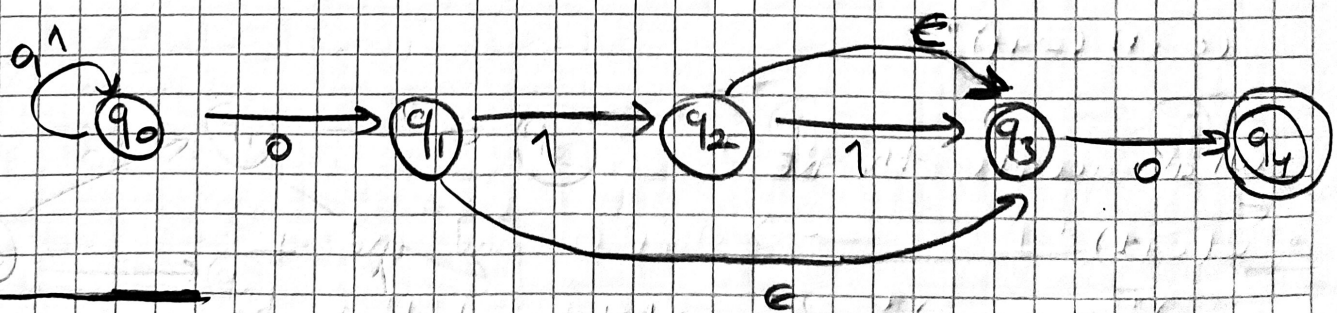
$\Sigma = \{0, 1\}$ NFA accepts $\{1, 111, 0, 011, 01, 000\}$



Transition Table

	0	1	ε
q_0	q_4	q_1	ϕ
q_1	ϕ	q_2	q_3
q_2	ϕ	q_3	ϕ
*q_3	ϕ	ϕ	ϕ
q_4	q_5	ϕ	ϕ
q_5	q_3	ϕ	ϕ

Ex:2 NFA $\Sigma = \{0, 1\}$ $L = \{w \mid w$ ends with 00, 010 and 0110$\}$



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Regular Expression

$(0+1) \rightarrow 0, 1$

or ← $01 \rightarrow 01$

$(01)^* \rightarrow 010101$

$(0+1)^* \rightarrow \epsilon, 0, 1, 00, 11, 01, 10$

Ex: RE

- all string of 0's, including the empty string
  $0^* \rightarrow \epsilon$ 'de gelebilir sonsuz sıfırda

- all strings of 0's and 1's without true consecutive 1's
  $(0+10)^*(\epsilon+1)$

- " length 2
  $(0+1)(0+1)$

- " including the empty string
  $(0+1)^*$

- " without empty string
  $(0+1)(0+1)^*$

- what accepts the RE?
- $0(0+1)^*1$ ⟶ start 0 and end 1
- $(0+1)^*00(0+1)^*$ ⟶ including at least 00
- $0^*(\epsilon+1)0^*$ ⟶ en fazla 1 tane 1 içeren (at most)
- $((1+01)^*(\epsilon+0))$ ⟶ ardışık iki sıfır içermeyen
- $(1+0)^*1(1+0)(1+0)$ ⟶ (2 basamaklı, ve sondan 2. hanesi 1

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- $1^*(01^*01^*)^*$ → even number zero (çift sayıda sıfır içeren)
- $0(1+0)$ → {01, 00}

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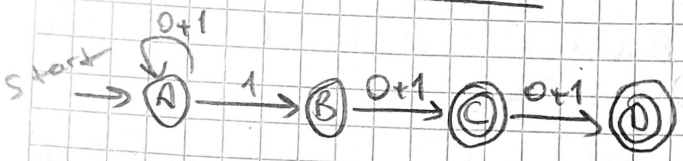
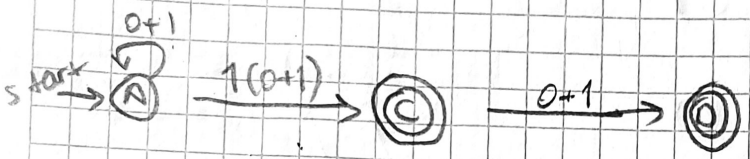
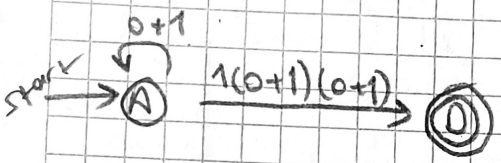
State Elimination

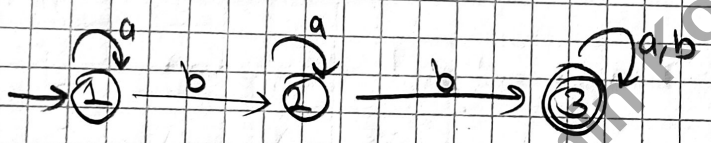






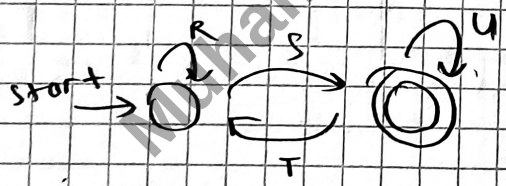
$$(0+1)^* 1 (0+1)(0+1) + (0+1)^* 1 (0+1)$$

Ex-2



$$a^* b a^* b (a+b)^*$$

Ex-3



$$(R^* + SU^* T)^* SU^*$$

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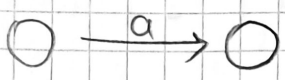
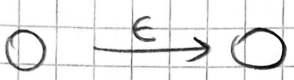
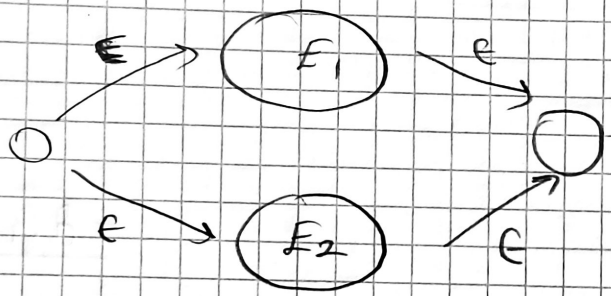
Converting a RE to an ϵ-NFA

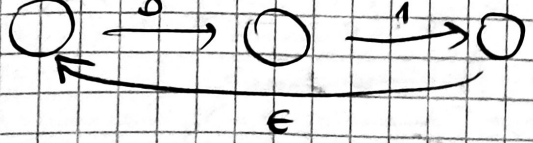
- Symbol a : 
- ϵ : 
- ∅ :

RE to ϵ-NFA Union



$E_1 \cup E_2$

$E_1 E_2$



$(01)^*$ without empty

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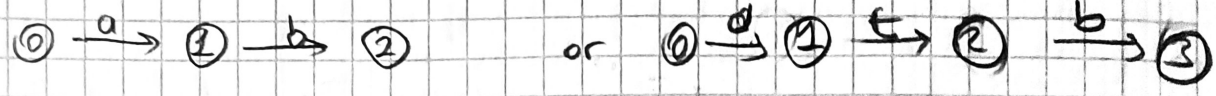
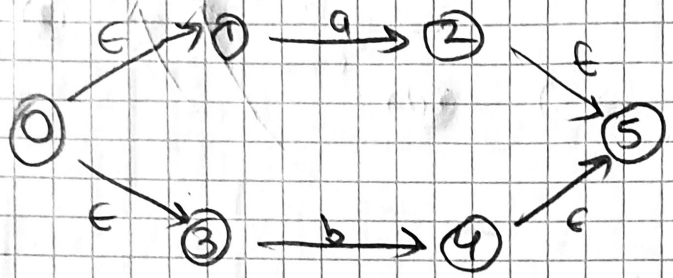
Ex : ab

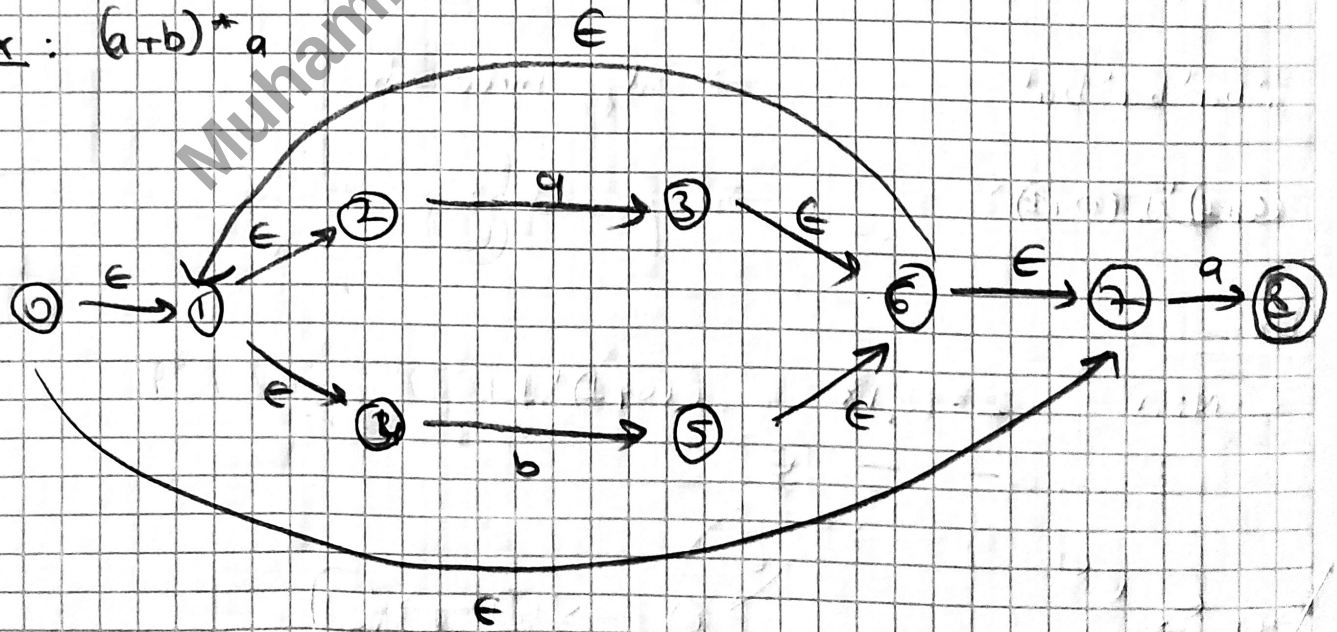


Ex : a+b



Ex : a^*

Ex : $(a+b)^* a$



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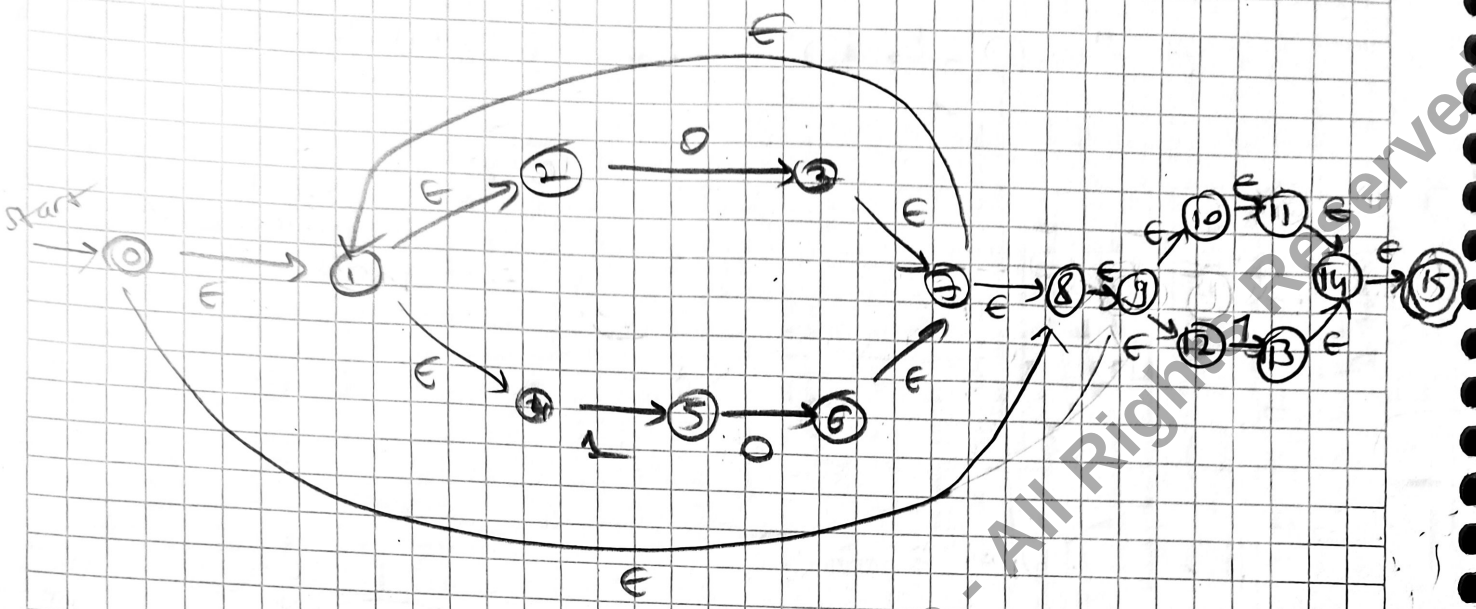
Ex : $(0+10)^*(\epsilon+1)$



LAB-5

$0(0+1)^*1 + 1(0+1)^*0$ → start and end different

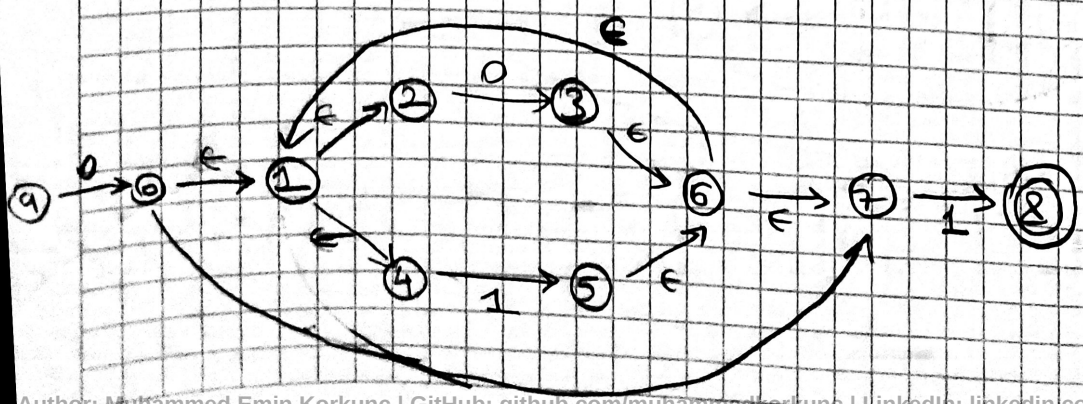
$0^*(1+\epsilon)0^*(1+\epsilon)0^*$ → at most two 1

$a^*ba^*ba^*ba^*$ → only three b

$(0+1)^*1(0+1)^*$ → at least 1

EX 2

ε-NFA $\quad \Sigma = \{0,1\}$ $\quad 0(0+1)^*1$



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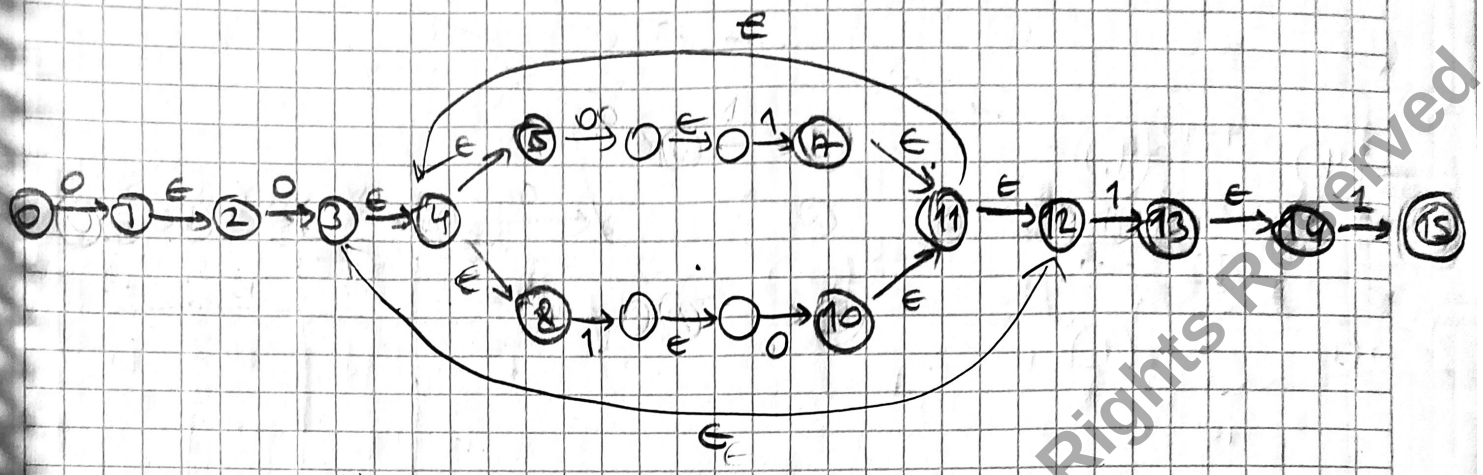
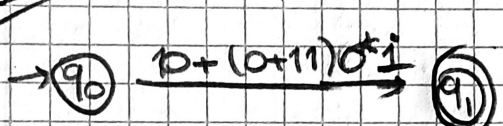
Ex3 : ϵ-NFA Σ={0,1} $00(01+10)^*11$

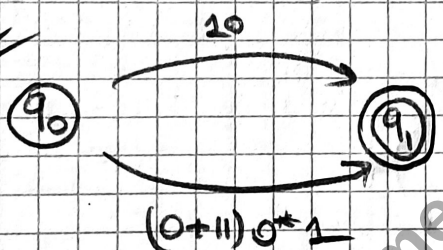


Ex4 : NFA Σ={0,1} $10+(0+11)0^*1$

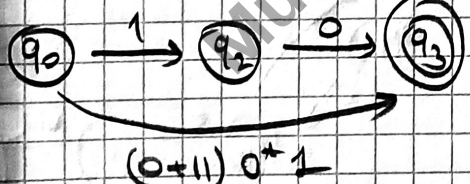
Step 1



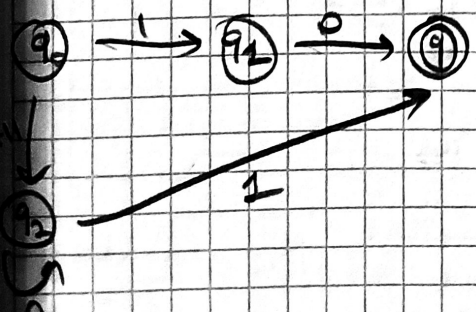
Step 2



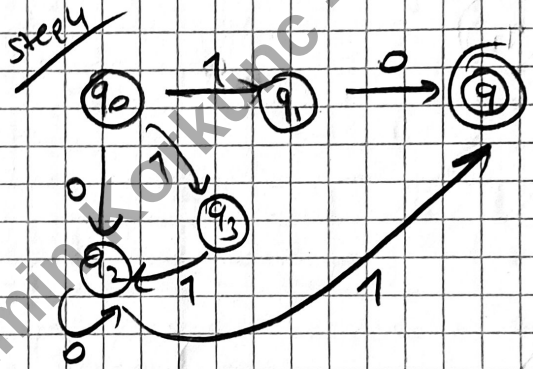
Step 3



Step 4



Step 4



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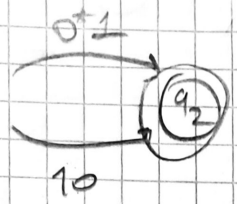
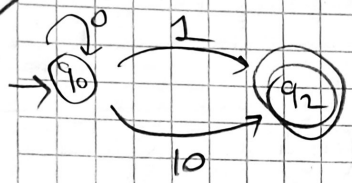
Ex5: NFA $\Sigma = \{0,1\}$ $0^*1 + 10$

Step 1

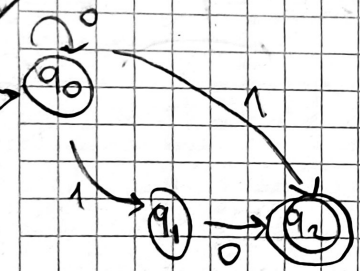
Step 2



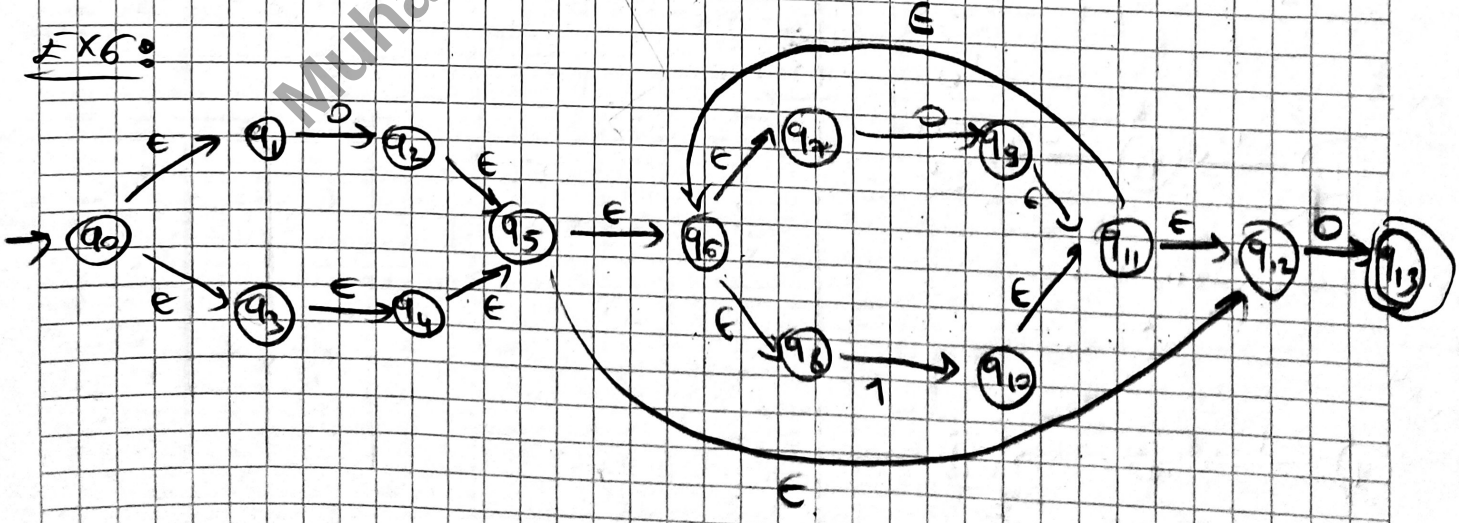
Step 3



Step 4



Ex6:



$(0+\epsilon)(0+1)^*0$

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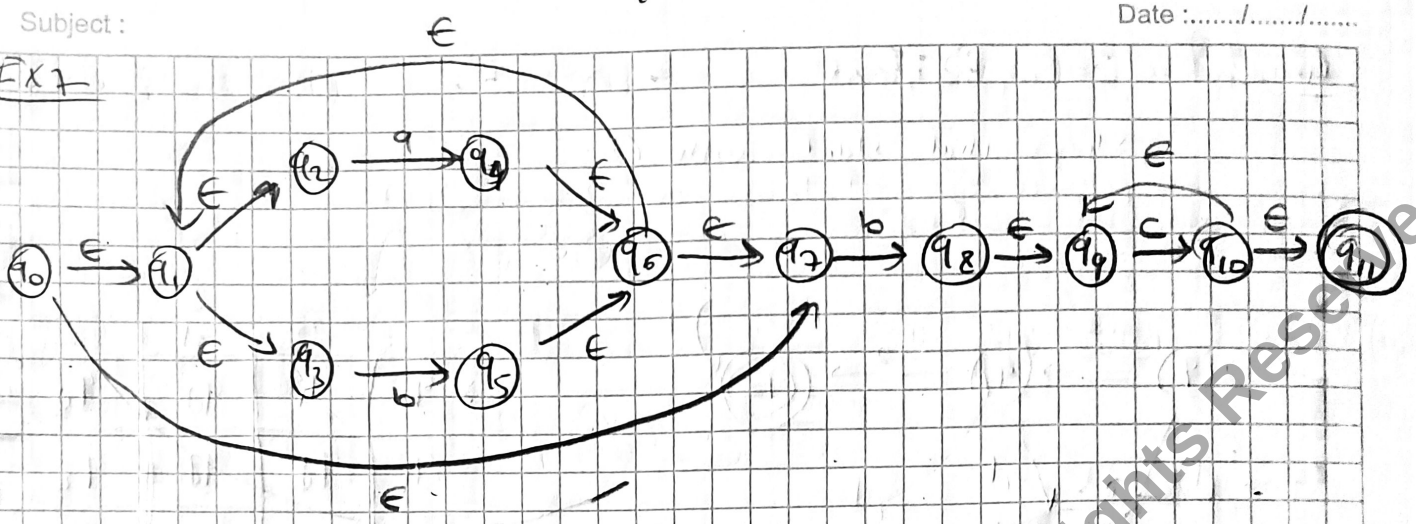
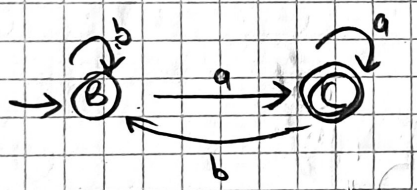
Ex1



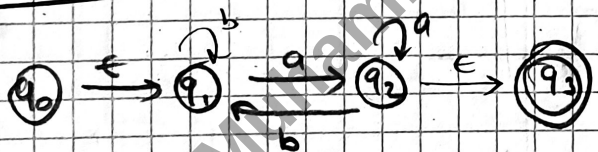
$(a+b)^* b c c^*$

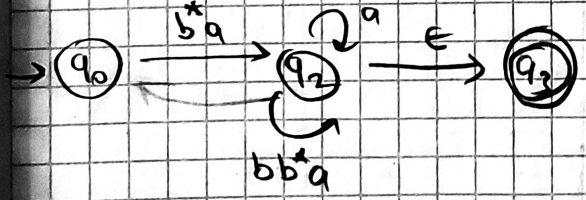
Write RE form this DFA

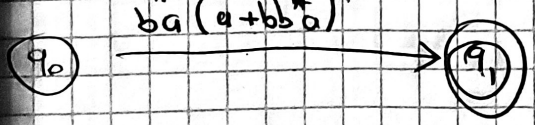


$(b^* + aa^* b)^* aa^*$ → Sabahan hoca

Çözüm







Zeliha hoca

$b^* a (a + b b^* a)^*$

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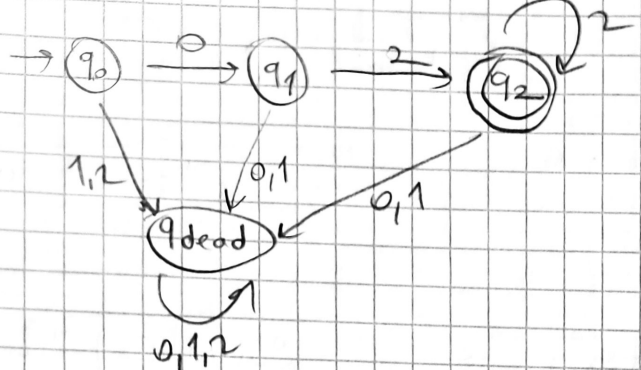
Quiz 1: $\Sigma = \{0, 1, 2\}$

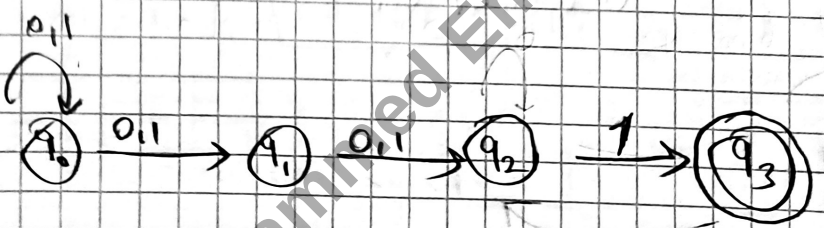
strings that start with 0 followed by at least one 2 DFA



	0	1	2
→ q_0	q_1	q_d	q_d
q_1	q_d	q_d	q_2
q_2	q_d	q_d	q_2
q_{dead}	q_d	q_d	q_d

Quiz 2: $\Sigma = \{0, 1\}$

end with 1
length at least 3 NFA



	0	1
→ q_0	q_0, q_1	q_0, q_1
q_1	q_2	q_2
q_2	$\emptyset$	q_3
* q_3	$\emptyset$	$\emptyset$

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Properties of Regular Language

1. Decision properties
2. Closure properties

Pigeonhole Principle

then there is at least one hole with more than one pigeon

Pumping Lemma for Regular languages

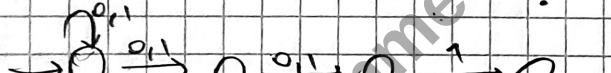
$w = 0^n 1^n$ → irregular

x	y	z
0	0	11

→ x y y z → 0 0 0 11

x	y	z
0	01	1

→ x y y z → 0 01 01 1



→ irregular (NFA)

x	y	z
0	0	1

→

	y	
0	00	1
0	000	1
0	0000	1

irregular

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Emptiness Problem

Infinite Cardinality

Q: Are all regular language finite?

Union Example

Pumping Lemma

$S = xyz$, where $|xy| \leq m$, $|y| > 0$

$\forall i \geq 0$, $xy^iz \in L$

EX

$L = \{$all palindromes over $\{a,b\}$ of odd length$\}$

palindrom → $S = S^R$ → S'nin tersi

$w = xyz$ $\quad |xy| \leq p \quad |y| \neq \epsilon \quad , |y| > 0$

$\forall i \quad xy^iz \in L$

$w = a^p b a^p$

$x = a^k$

$y = a^m$

$z = a^{p-k-m} b a^p$

$k+m \leq p \quad m > 0$

$\forall i \; xy^iz \in L$

$i=0$ için xz

$i=1$ için xyz

$i=2$ için $xyyz = xy^2z$

↳ $a^k a^{2m} a^{p-k-m} b a^p = \boxed{a^{p+m} b a^p}$ → son ifade

$\neq \boxed{a^p b a^{p+m}}$ → son ifadenin tersi: kendisine eşit olmadığı için palindrom değil yani

Not regular

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Ex: $L = \{0^i 1^j \mid i, j \in N \; i > j\}$

$w = xyz \rightarrow 0^{p+1}1^p \rightarrow \underbrace{000\ldots0}_{p+1}\underbrace{1\ldots1}_{p}$ $\quad |xy| \leq p \quad |y| > 0$

$x = 0^a$
$y = 0^b 1^c$
$z = 0^{p+1-a-b} 1^{p-c}$

$a+b \leq p+1$ ilerletyemem

$x = 0^a$
$y = 0^b$
$z = 0^{p+1-a-b} 1^p$

$\forall i \; xy^i z$
$i=0$ için xz
$0^a 0^{p+1-a-b} 1^p$

$0^{p+1-b} \rightarrow p+1-b > p$
$1 > b$

$|y| > 0$ Not regular

Ex3 Prove that $L = \{a^n b^n c^n, n \geq 0\}$ is not a regular language for xy^2z

$w = xyz$ p
$x = a^k$
$y = a^m$
$z = a^{p-k-m} b^p c^p$

$xyyz$
$a^k a^{2m} a^{p-k-m} b^p c^p$
$= a^{p+m} b^p c^p$ not regular

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Ex4 : Prove that $L=\{ww : w \text{ in } \{a,b\}^*\}$ is not a regular language

$ww \rightarrow ab|ab$

$a|a$

ax

$ww \rightarrow \underbrace{a^p b}_{w} \underbrace{a^p b}_{w} = w$

$a^p(1+b)$

$w = xyz$

$x = a^k$

$y = a^m$

$z = a^{p-k-m} b a^p b$

$i=2$ için $xyyz$

$a^{p+m} b a^p b$ → not regular

Çünkü ifade bozuldu $a^p b a^p b$ diye gitmesi lazımdı, fakat → m adın sayısını bozuyor

Ex5: $L=\{a^n b^t \mid n,t \in N, n \leq t\}$

$w = a^p b^{p+1}$

$x = a^k$

$y = a^m$

$z = a^{p-k-m} b^{p+1}$

$i=2$ için $xyyz$

$a^k a^{2m} a^{p-k-m} b^{p+1}$

$a^{p+m} b^{p+1}$ → not regular

$i=3$ için $xyyyz$

$a^{p+2m} b^{p+1}$

$w = aa^b bbb$

$x = a$

$y = a$

$z = bbb$

$x=0$ için xz abbb ✓

$i=1$ için xyz aabbb ✓

$i=2$ için xy^2z aaabbb ✓

$i=3$ için xy^3z aaaabbb ✗ not regular

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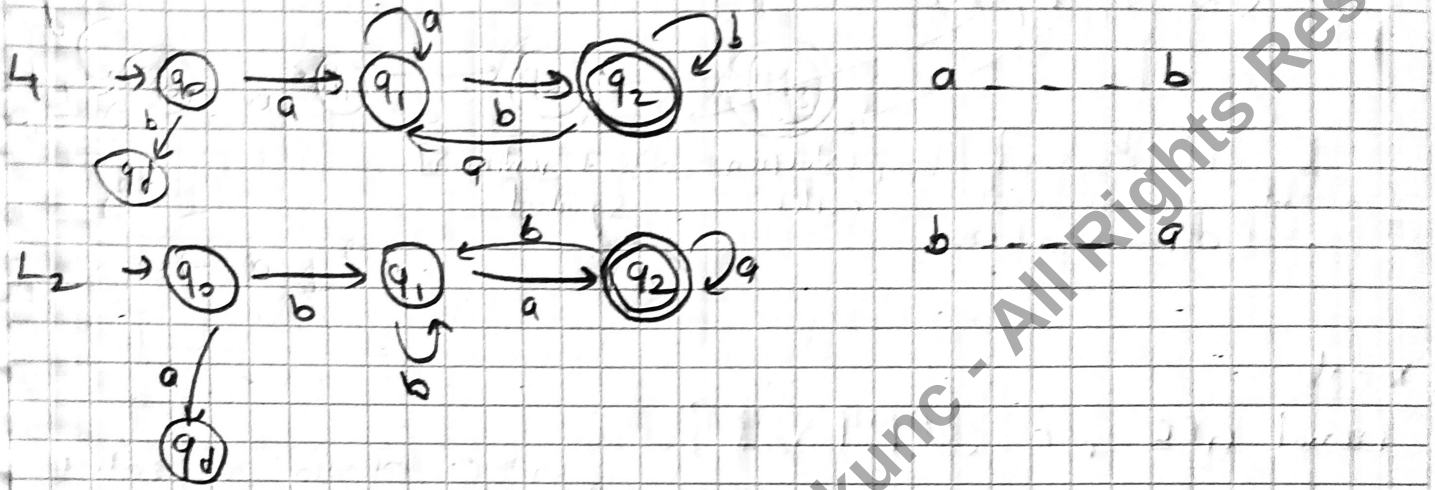
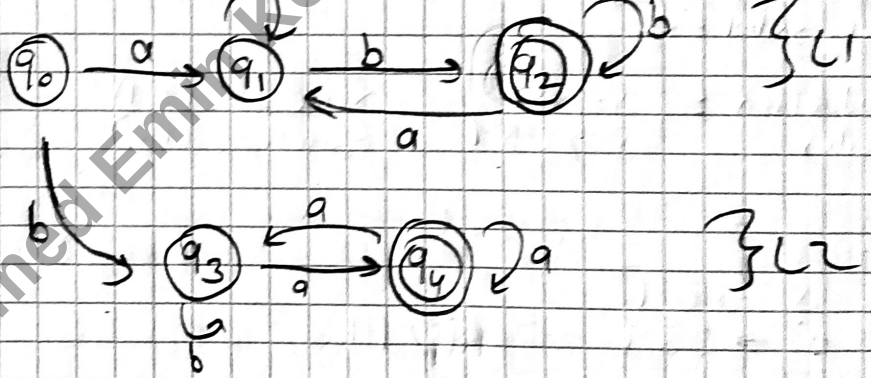
Ex 6 / L_1 = {a ile başlar b ile biter}

L_2 = {b ile başlar ve a ile biter}

DFA for $L = L_1 \cup L_2$ or $L = L_1 + L_2$



$L_1 \cup L_2$ $(L_1 + L_2)$



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Formal Language and Automata Theory

Context Free Grammers

$G = (V, T, P, S)$

- V: nonterminal / variable symbol
- T: terminals / terminal symbol
- P: production rules
- S: start nonterminal symbol

Example

Formal GFC for $\{0^n 1^n \mid n \geq 1\}$ → $n \geq 0$ olursa $\in$ kabul edebilir

Terminals = $\{0,1\}$
Variables = $\{S\}$
Start symbol = S

Production rules = $S \to 01$ Rule 1 (R1)
$S \to 0S1$ Rule 2 (R2)

$\to S \to 01$
$S \to 0S1 \to 0011$

000111 yap → $S \xrightarrow{R_2} 0S1 \xrightarrow{R_2} 00S11 \xrightarrow{R_1} 000111$

$0S1$: $S \to 0S1$; $00S11$: $S \to 01$

Eğer Kural şu şekilde olsaydı.. Rule 1 ⇒ $S \to 0S1$
Rule 2 ⇒ $S \to \epsilon$

000111 yapmak için → $S \to 0S1 \xrightarrow{R_1} 00S11 \xrightarrow{R_1} 000S111 \xrightarrow{R_2} 000111$

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Palindrome Example

A: $V = \{x\}$

$\Sigma = \{0, 1\}$

$R = \{x \to \epsilon\ (R_1),\ x \to 0\ (R_2),\ x \to 1\ (R_3),\ x \to 0x0\ (R_4),\ x \to 1x1\ (R_5)\}$ ← rule

$S = x$

Standardize pal, reset

$V = \{S\},\ S = S$

$R = \{S \to \epsilon,\ S \to 0,\ S \to 1,\ S \to 0S0,\ S \to 1S1\}$

ϵ ; $x \xrightarrow{R_1} \epsilon$

1 ; $x \xrightarrow{R_3} 1$

010 ; $x \xrightarrow{R_4} 0x0 \xrightarrow{R_3} 010$

1011101 ; $x \xrightarrow{R_5} 1x1 \xrightarrow{R_4} 10x01 \xrightarrow{R_5} 101x101 \xrightarrow{R_3} 1011101$

R_5, R_4, R_5, R_3

0000 ; $x \xrightarrow{R_4} 0x0 \xrightarrow{R_4} 00x00 \xrightarrow{R_1} 0000$

R_4, R_4, R_1

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Kleene closure

Grammer for unsigned integers can be replaced by

$U \to UD \mid D$

$D \to 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

Example: $S \to 0S1 \mid 01$ is shorthand for

$S \to 0S1$
$S \to 01$

10 ; $U \to UD \to DD \to 1D \to 10$

$U \to UD \to UD \to DD \to 10$

Ex: Leftmost Derivations

- Balanced parentheses grammar

$S \to SS \mid (S) \mid ()$

- $S \to_{lm} SS \to_{lm} (S)S \to_{lm} (())S \to_{lm} (())()$

Thus, $S \to^*_{lm} (())()$

- $S \to SS \to S() \to (S)() \to (())()$

Ex: Rightmost Derivations

$S \to SS \mid (S) \mid ()$

$S \to_{rm} SS \to_{rm} S() \to_{rm} (S)() \to_{rm} (())()$

Thus $S \to^*_{rm} (())()$

$S \to SS \to SSS \to S()S \to ()()S \to ()()()$

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More Examples

The grammar ({A}, {a,b,c}, P, A)

A → aA

A → abc

Çözüm →

A → abc
A → aA → aabc
A → aA → aaA → aaabc

} {abc, aabc, aaabc, ---

- The grammar ({S}, {a,b}, P, S)

S → aSa
S → bSb
S → ϵ

→ Büyük harfle yazılan → Terminal
→ Küçük harfle yazılan → Variable

- The grammar ({S,F}, {0,1}, P, S)

S → 00S | 11F
F → 00F | ϵ

Construct the CFG for the language having any number of a's over the set Σ = {a}

Sol: S → aS → ϵ

Construct a CFG for the regular expression (0+1)*

Soln: S → 0S | 1S | ϵ

Construct a CFG for the language L = a^n b^2n where n >= 1

Sol: S → aSbb | abb

A CFG for all binary strings with an even number of 0's

Sol: S → 1S | 0A0S | ϵ
A → 1A | ϵ

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Ex1 : $G = (\{S, A\}, \{0, 1\}, P, S)$

$S \to 1S \mid 0A0S \mid \epsilon$

$A \to 1A \mid \epsilon$

Show derivations for 101011

101011 ; $S \to 1S \to 10A0S \to 101A0S \to 1010S \to 10101S$

$\to 1010115 \to 101011$

Ex2 : $G = (V, T, S, P)$

P =

$S \to ABC$

$A \to aA \mid \epsilon$

$B \to bB \mid \epsilon$

$C \to cC \mid \epsilon$

Show derivations for 'abbc'

abbc ; $S \to ABC \to aABC \to aBC \to abBC \to abbBC \to$

Left most derivations $abbC \to abbcC \to abbc$

abbc ; $S \to ABC \to ABcC \to ABc \to AbBc \to AbbBc \to$

Right Most derivations $Abbc \to aAbbc \to abbc$

Ex3 : Grammar for if | else $V = (i, e)$

- if if else
- if if else else

Cevap

$S \to iS \mid iSe \mid \epsilon$

if if else

$S \to iS \to iiSe \to iie$

if if else else

$S \to iSe \to iiSee \to iiee$

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Ex4: Grammer for Addition / Subtraction

$G = (V, T, P, S)$ $V = E$ $T = \{1, 0, +, -\}$ $S = E$

Show derivations for '0+0-1'

0+-1 üretilir mi?

Çözüm; $E \rightarrow E+E \mid E-E$ → Grammer biz yazdık

$E \rightarrow 0 \mid 1$

0+0-1 ; $E \rightarrow E+E \rightarrow E+E-E \rightarrow 0+E-E \rightarrow 0+0-E \rightarrow 0+0-1$

0+-1 ; üretilemez

Ex5: Constructor a CFG for the language $L = a^n b^{2n}$ where $n >= 1$

Run for aaabbbbbb

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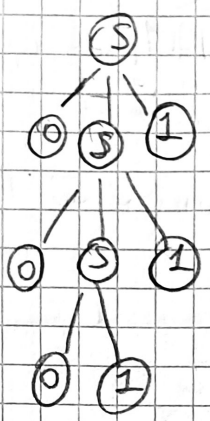
Parse Trees

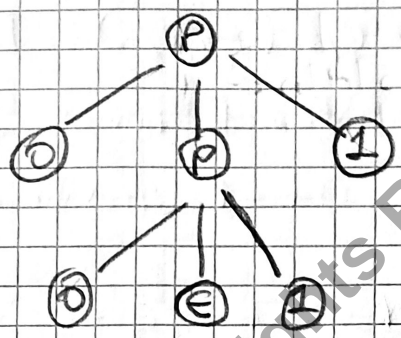
Ex 2

S -> 01
S -> 0S1

000111



0011



P -> 0P1
P -> ε

S -> 0S1 -> 00S11 -> 000111

Yield (Getiri) of a Parse Tree

Ambiguous Grammars

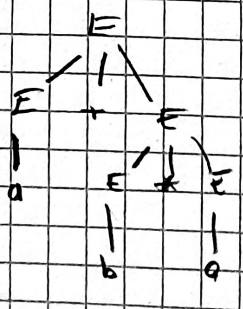
iki farklı parse tree tanımlanabiliyorsa ambiguous'dur.

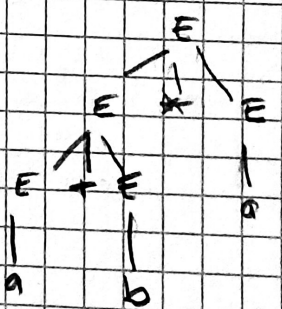
Ex 3

A context free grammar for simple expression symbol {+, *, (,), a, b, 0, 1}

1. E -> I
2. E -> E+E
3. E -> E*E
4. E -> (E)
5. I -> a
6. I -> b
7. I -> Ia
8. I -> Ib
9. I -> I0
10. I -> I1

a+b*a





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2 (1) Grammars

Ex 1

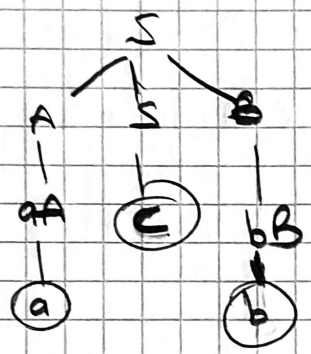
$S \to ASB \mid c$

$A \to \varepsilon \mid aA$

$B \to \varepsilon \mid bb$

acb 'yi elde etice-

Cevap → $S \to ASB \to a\,ASB \to a\,SB \to a\,cB \to acbB \to acb$

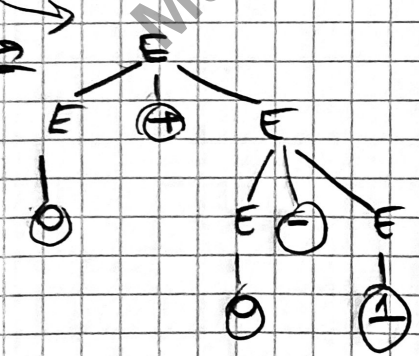


Ex 2: grammar for Addition / Subtraction

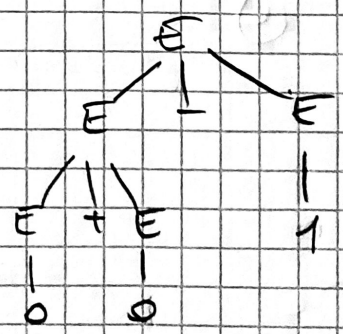
$G = (V, T, P, S)$ $V = E$ $T = \{1, 0, +, -\}$ $S = E$

$E \to E+E \mid E-E$
$E \to 0 \mid 1$
$0+0-1$

Cevap →



Ambiguous



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Ex 3 :

① $S \to a|b|c|SRS$

$R \to +|-|*|/$

② $S \to L|LRS$

$L \to a|b|c$

$R \to +|-|*|/$

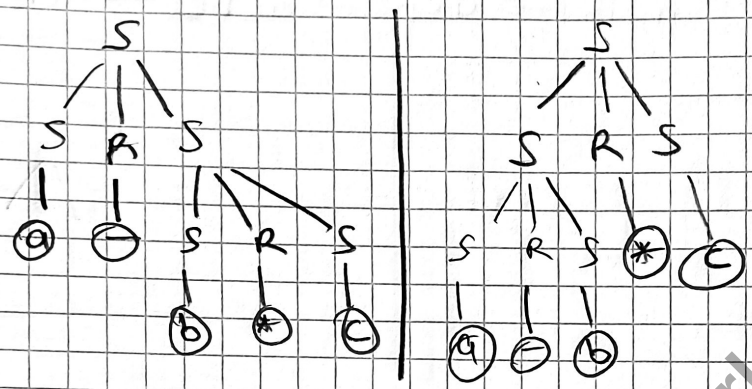
③ $S \to L|SRL$

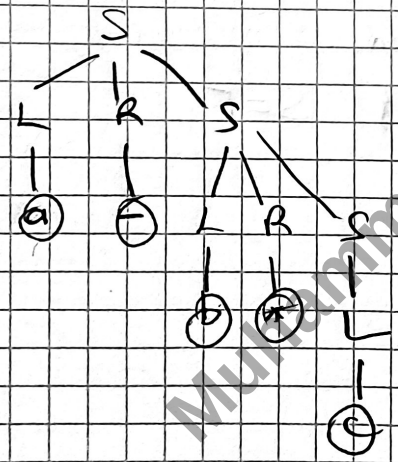
$L \to a|b|c$

$R \to +|-|*|/$

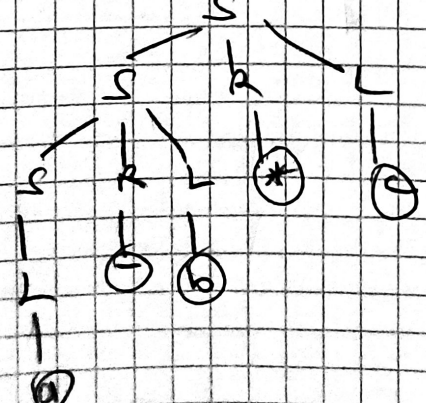
a - b * c buna bul 3 ağaç ile

Çözüm

① 

② 

sola yorlama

③ 

sağa yorlama

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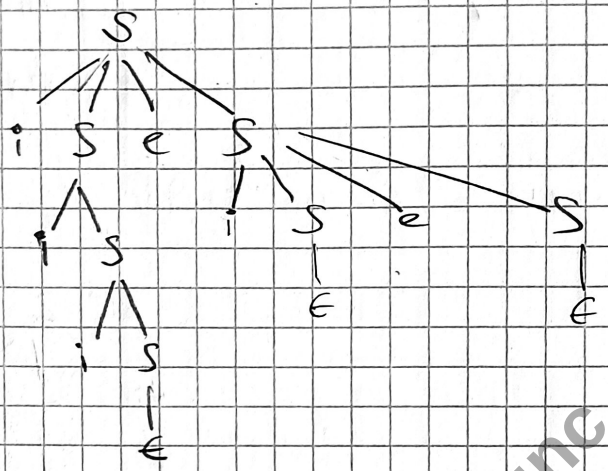
Subject :

Date :...../...../.....

Ex4 : $S \to \epsilon \mid SS \mid iS \mid iSeS$

iiieie

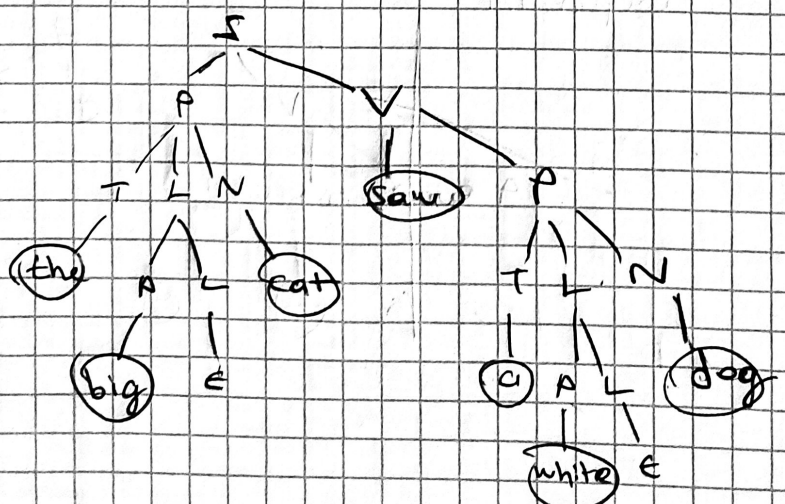
Çözüm →



EX

$S \to PV$
$P \to TLN$
$L \to \epsilon \mid AL$
$T \to \epsilon \mid the \mid a$
$A \to big \mid white \mid lazy$
$N \to cat \mid dog \mid bug$
$V \to ran \mid saw \mid ate$ P

The big cat saw a white dog



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